

Queen Mary University of London Engineering School, NPU

Queen Mary University of London

QXU6032
Module Name: Advanced Ceramic and Glass Materials

Group coursework (4 students) 2026

Detailed technical report on the state of the art of an advanced ceramic material: Zirconia Toughened Alumina Ceramic Femoral Head

Complete the information below about yourself very carefully.

Group name: Alumina5

DECLARATION

Zirconia Toughened Alumina Ceramic Femoral Head

Was composed by our group and is based on our own work. Where the work of the others has been used, it is fully acknowledged in the text and in captions to table illustrations. We also declare that all information in this report has been acquired and presented in accordance with academic rules and ethical conduct of QMES. The use of AI editing tools should be approved by the module organizer prior to submission. AI Usage Log listing each AI tool used, the purpose, and the extent of its contribution. **Failure to disclose AI use is treated as academic misconduct.** The module organizer reserves the right to hold oral exams for any student to verify understanding of their contributed section. Each individual student must be familiar with the contents of the entire report.

Public study copy: student numbers and handwritten signatures removed.
The group contribution statement and AI disclosure are retained.

Examiner

Dr Salvatore Grasso

Copyright © Northwestern Polytechnical University & © Queen Mary University of London 2026

Task 1	Select an application and the matching reference ceramic composition. For example, “dental zirconia” (10 marks/100)	
Suggested length	Indicative length 500 words	**Xuanming Che、Haotai Xu、Xiangjun Zheng、Yongqi Xu**

1.1 Introduction and Background

Total hip arthroplasty is a key treatment for end-stage hip disease. A typical total hip prosthesis system consists of a femoral stem, femoral head, acetabular cup, and liner, with the femoral head and liner forming the primary load-bearing and sliding interface. This interface is subjected to long-term cyclic contact loads, frictional shear stresses, and a complex environment of physiological fluids.[1] Over the past decade, the average age of patients undergoing this surgery has gradually decreased, which requires artificial joint materials to maintain these properties over a longer service life.[2]

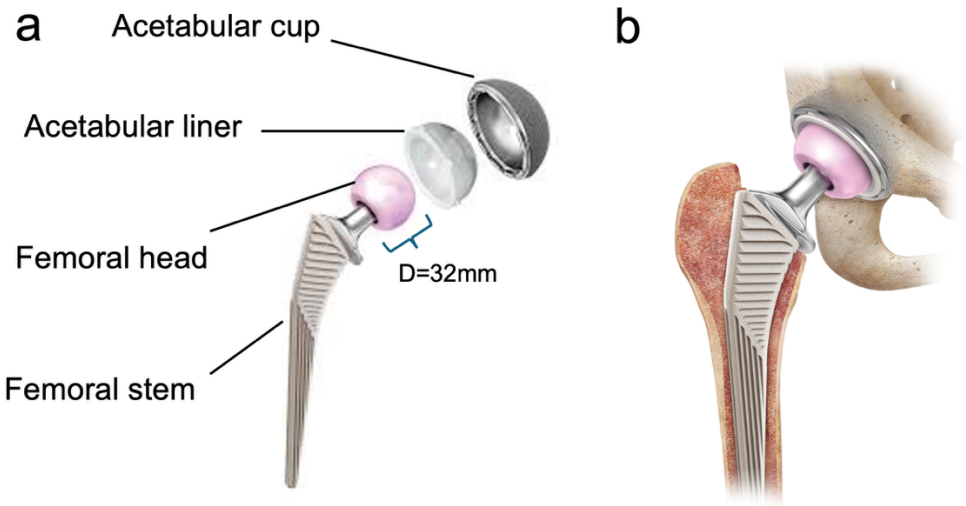


Figure 1-1-1. Schematic diagram of the main components and assembly of a hip replacement system. (a) Exploded view of the artificial hip joint prosthesis, including the acetabular cup, acetabular liner, femoral head, and femoral stem. (b) The functional state of the artificial hip joint prosthesis after implantation.[3]

Alumina ceramics were first successfully used in artificial hip replacement by Boutin in 1970. [4]. To date, more than 3 million alumina ceramic ball joints have been installed worldwide. [5]. Ceramic femoral heads have become an important material choice for the load-bearing pair in total hip replacements.

The mechanical operating conditions for ceramic femoral heads are extremely demanding. ISO 14242-1:2014 provides a standardized testing environment for wear testing of total hip prostheses[6], used to simulate the friction and wear behavior of the femoral head-liner interface during long-term joint movement. During high-load activities such as walking up stairs, concentrated Hertzian contact stresses are generated at the femoral head-liner contact interface[7], approximately 2.5 to 3 times their body weight [8]. Even more dangerous are the tensile stresses that may develop at the edges, on surface defects, and near the transition zone of the femoral head’s internal taper; these locations could serve as initiation points for slow crack propagation or catastrophic fracture.

Based on the aforementioned service conditions and in accordance with the material-level requirements for alumina-based composite ceramics for surgical implants

specified in ISO 6474-2:2019, this report adopts the following criteria as the evaluation standards for ceramic femoral heads [9]

Table 1-1-1. Key performance requirements and target values for ceramic hip joint materials.

Performance Requirements	Target
Relative density	≥99%
4-point bending strength	≥750–1000 MPa
Fracture toughness	≥3.5–4.0 MPa·$m^{1/2}$
Vickers hardness	≥15.5–16.0 GPa

This report selects zirconia-toughened alumina (ZTA) as a candidate material to meet the aforementioned service requirements. This system uses α-Al_2O_3 as the continuous matrix, providing high hardness, low wear rates, and biocompatibility; Y_2O_3-stabilized metastable tetragonal ZrO_2 particles serve as the dispersive toughening phase, undergoing a t-ZrO_2 → m-ZrO_2 phase transformation under stress at the crack tip, thereby inhibiting crack propagation through localized volume expansion and the creation of a compressive stress field. [10,11] The ZrO_2 content examined in this report ranges from 10% to 20% by weight, with Y_2O_3 accounting for 3% of the ZrO_2.

This report will subsequently discuss, in turn, its crystal structure and bonding, phase stability, fabrication process, and final performance characterization.

1.2 Justification for the Chosen Application

ZTA ceramic femoral heads have clear clinical significance. Total hip replacements are intended for long-term use within the body, and the risk of revision accumulates over time. Data in Figure 1-2-1 show that the long-term cumulative revision rate is higher in younger patients than in older patients, with male patients under 55 years of age having a revision rate as high as 11.18% at 20 years. This indicates that younger, more active patients have a greater need for bearing materials with low wear and high reliability.

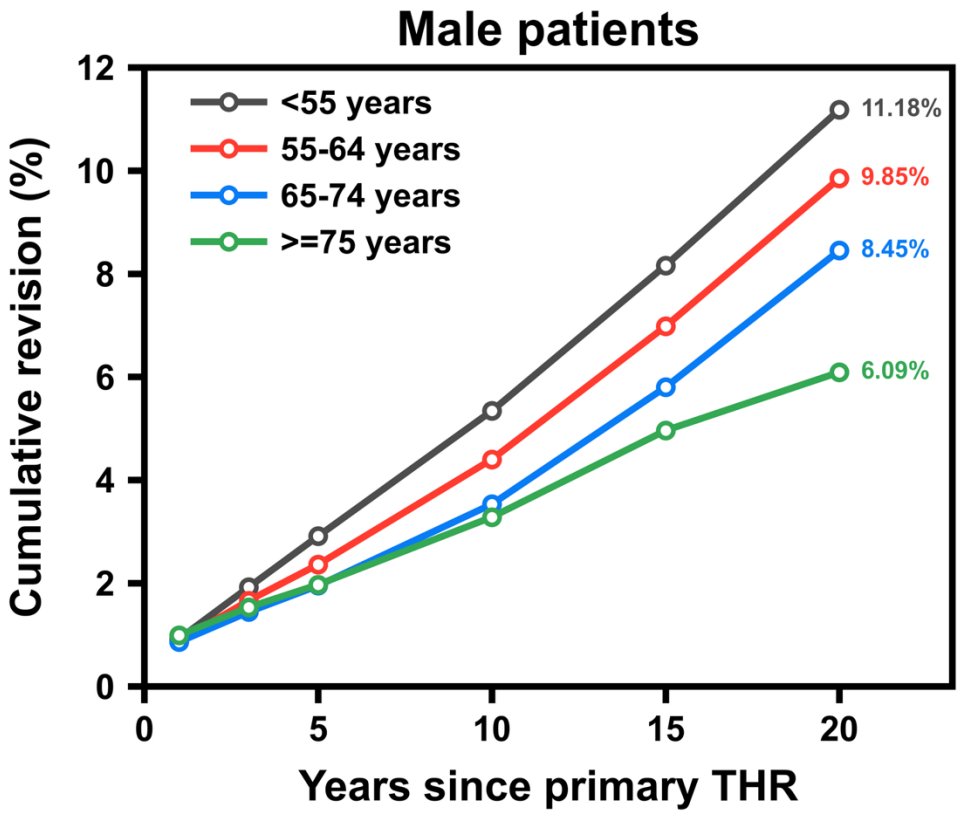


Figure 1-2-1. Estimated cumulative revision rates for male patients following primary total hip arthroplasty. Highest estimated cumulative revision rate. (This figure was created by us based on data extracted from Table 3.H6 of the NJR 22nd Annual Report 2025.)[12]

The data in Figure 1-2-2 show that the long-term cumulative revision rate for ceramic-on-ceramic (CoC) bearings is significantly lower than that for metal-on-metal (MoM) bearings, and is comparable to that for metal-on-polyethylene (MoP) bearings, indicating that ceramic bearings have potential clinical value in long-term total hip arthroplasty (THA) applications.

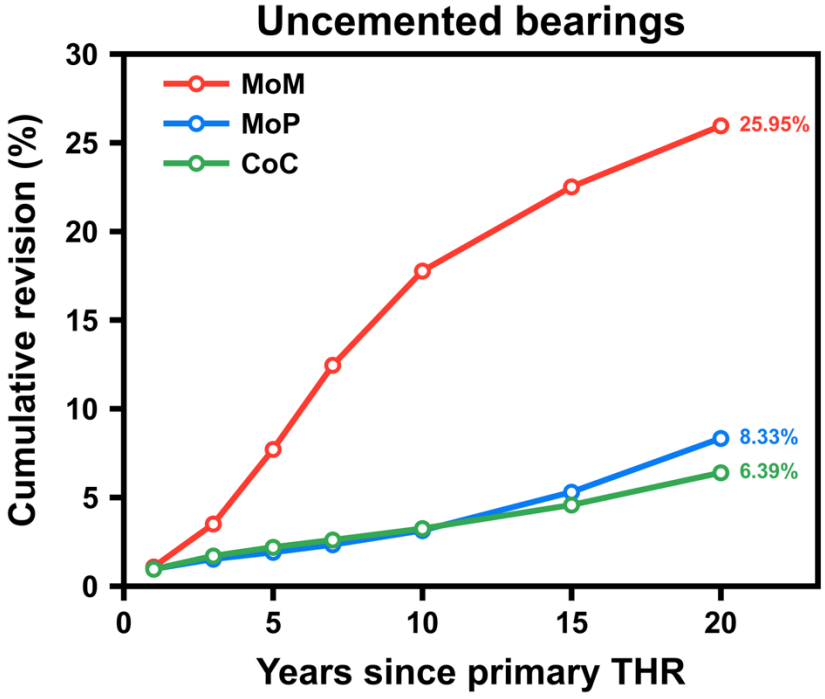


Figure 1-2-2. Estimated cumulative revision rates for different bearing surfaces in primary cementless total hip arthroplasty. (This figure was created by us based on data extracted from Table 3.H5 of the NJR 22nd Annual Report 2025.) [12]

The Ashby-style diagram in Figure 1-2-3 is used to further illustrate the advantages of ZTA in material selection. Although engineering polymer materials are closer to the modulus of bone tissue, they lack sufficient strength, hardness, and wear resistance. Metallic materials such as Ti alloys, Co–Cr alloys, and stainless steel possess high fracture toughness, but their surfaces are susceptible to corrosion and the release of metal ions.

In contrast, ceramic materials are concentrated in the region of high elastic modulus and moderate fracture toughness. Single-phase Al_2O_3 ceramics possess high elastic modulus and excellent hardness, but their fracture toughness is typically low, making them sensitive to porosity, scratches, and microcracks. Stress-induced phase transformations in single-phase ZrO_2 ceramics can yield higher fracture toughness, but hydrothermal degradation at low temperatures may occur in humid environments. ZTA lies between the performance regions of Al_2O_3 and ZrO_2, combining the wear resistance of alumina with the phase-transformation toughening capability of zirconia, thereby achieving a better balance between low wear, fracture resistance, and long-term stability.

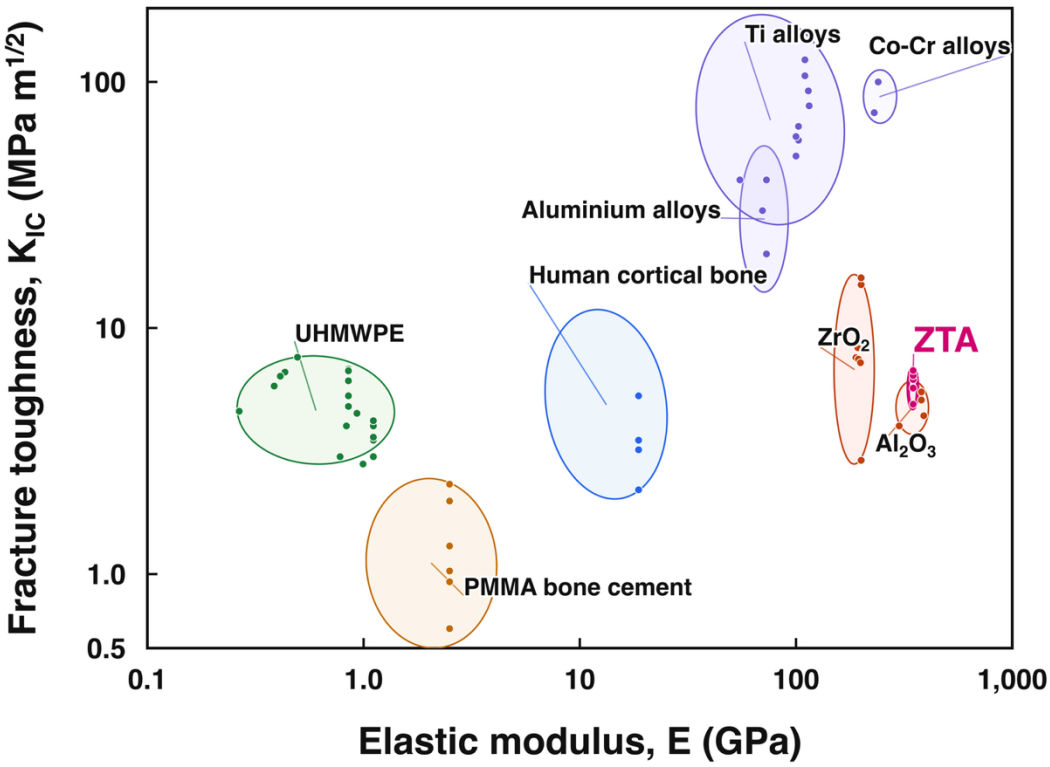


Figure 1-2-3. Ashby diagram comparing the elastic modulus E and fracture toughness K_{IC} of hip joint bearing materials. The colored ellipses indicate the approximate performance ranges of human cortical bone, engineering polymers, metal alloys, and

oxide ceramics. (This figure was created by us using data extracted from the literature.)[13–20]

Task 2	Crystal structure, atomic bonding, and polymorphs (20 marks)	
Suggested length	Indicative length 1000-1500 words	**Haotai Xu**

ZTA is a biphasic composite bioceramic system consisting of α-Al_2O_3 (corundum) as the continuous matrix phase and Y_2O_3-stabilized tetragonal zirconia (t-ZrO_2) as the dispersed toughening phase. Both phases exhibit a hybrid bonding characterization characterized by the coexistence of ionic and covalent bonds, This core bonding property determines the high mechanical stability, chemical inertness, and long-term biocompatibility of ZTA ceramics in physiological environments.

2.1 Atomic Bonding

The atomic bonding characteristics of the constituent phases of ZTA can be quantitatively characterized using Pauling's electronegativity theory. The formula for calculating the ionic fraction is:

$$f_i = 1 - e^{-0.25(\chi_A - \chi_B)^2} \quad (1)$$

For α-Al_2O_3, $\chi_O = 3.44$, $\chi_{Al} = 1.61$, substituting into the formula gives:

$$f_i(Al-O) = 1 - e^{-0.25\times(3.44-1.61)^2} \approx 0.567 \quad (2)$$

The Al-O bond is 56.7% ionic and 43.3% covalent.[21] This hybrid bonding characteristic, combined with a high bond energy of approximately 512 kJ/mol, endows α-Al_2O_3 with extremely high hardness (~2000 HV), a high melting point of 2072°C, and excellent chemical stability; its wide bandgap of 8.8 eV provides good electrical insulation and an extremely low ion elution rate, meeting the biocompatibility requirements for medical implant materials. [22]

For t-ZrO_2, $\chi_{Zr} = 1.33$, substituting into the formula gives:

$$f_i(Zr-O) = 1 - e^{-0.25\times(3.44-1.33)^2} \approx 0.671 \quad (3)$$

The ionic contribution of the Zr-O bond is 67.1%, while the covalent contribution is 32.9%; the ionic nature is significantly higher than that of the Al-O bond. [23] The higher ionicity of Y_2O_3 provides the structural basis for the introduction of oxygen vacancies and is also a key prerequisite for the stress-induced t→m phase transition; its bond energy of approximately 760 kJ/mol ensures structural integrity during the phase transition. [24]

For Y_2O_3, $\chi_Y = 1.22$, substituting into the formula gives:

$$f_i(Y-O) = 1 - e^{-0.25\times(3.44-1.22)^2} \approx 0.708 \quad (4)$$

The Y-O bond has the highest ionic contribution, at 70.8%[25] When Y^{3+} substitutes for Zr^{4+} in the ZrO_2 lattice, an oxygen vacancy is created due to the charge imbalance. This defect is represented by the Kröger-Vink notation as:

$$Y_2O_3 \xrightarrow{ZrO_2} 2Y'_{Zr} + V_{\ddot{O}} + 3O_O^{\times} \quad (5)$$

In the equation, Y'_{Zr} prime denotes a Y^{3+} ion occupying a Zr^{4+} lattice site with an effective charge of -1, and $V_O^{\cdot\cdot}$ denotes an oxygen vacancy with an effective positive charge of +2.[25] This defect reduces the free energy of the high-temperature tetragonal phase of ZrO_2, allowing the tetragonal phase to remain in a metastable state at room temperature.

Al_2O_3 and ZrO_2 exhibit good bonding compatibility, and the interface is clean with no harmful byproducts. [27] The formation energy of oxygen vacancies in ZrO_2 (~4.5 eV) is lower than that in Al_2O_3 (~6.0 eV), ensuring that Y_2O_3 doping selectively introduces oxygen vacancies into the ZrO_2 phase. [28]

The key performance differences between the two core compositions, derived from their bonding characteristics, are shown in Table 2-1-1.

Table 2-1-1. Key Performance Characteristics Derived from Bonding Features

Performance	α-Al_2O_3	t-ZrO_2	Reason
Ionic characteristics	56.7%[29]	67.1%[23]	Electronegativity difference
Coordination number	6(octahedron)	8(Twisted Cube)	Ion radius ratio
Bandgap(eV)	8.8[22]	5-7	O 2p-metal nd hybridization
Hardness(HV)	~2000	~1200	Bond strength and bulk density
Fracture toughness(MPa·m¹/²)	3-4	6-10(including Y_2O_3)	Phase change capability
Thermal conductivity(W/m·K)	30-35	2-3	Scattering of phonons by oxygen vacancies
Water-soluble	Negligible	Negligible	High lattice energy, strong M-O bond

The Al-O bonds form a rigid three-dimensional AlO_6 octahedral network due to their covalent character; the higher ionic character of the Zr-O bonds confers deformability to the lattice, enabling the transformation of the fluorite-derived structure.

These ionic differences can be further visualized using Morse potential curves. [30]

$$V(r) = D_e\left[1 - e^{-a(r-r_e)}\right]^2 \quad (6)$$

Here, De represents the dissociation energy (potential well depth, eV), re represents the equilibrium bond length (Å), and a represents the potential well width parameter (1/Å).

Table 2-1-2. Key Performance Characteristics Derived from Bonding Features.[30–33]

Atomic pairs	D_e (eV)	α ($Å^{-1}$)	r_e (Å)
Al–O	5.31	1.85	1.71
Zr–O	7.82	1.52	2.09
Y–O	7.41	1.40	2.25
Al–Al	1.80	1.20	2.86
Zr–Zr	1.60	1.05	3.45
O–O	5.17	2.65	1.21

As shown in Figure 2-1-1, the Zr–O bond potential well is the deepest (De = 7.82 eV), corresponding to the long-range Coulombic attraction resulting from its high ionic character of 67.1%; The Al–O potential well is the narrowest (a = 1.85 $Å^{-1}$), reflecting the strong directionality and high force constant resulting from its 43.3% covalency; the Y–O potential well is wide and shallow (a = 1.40 $Å^{-1}$), consistent with its highest ionicity of 70.8% and longest bond length (2.25 Å).

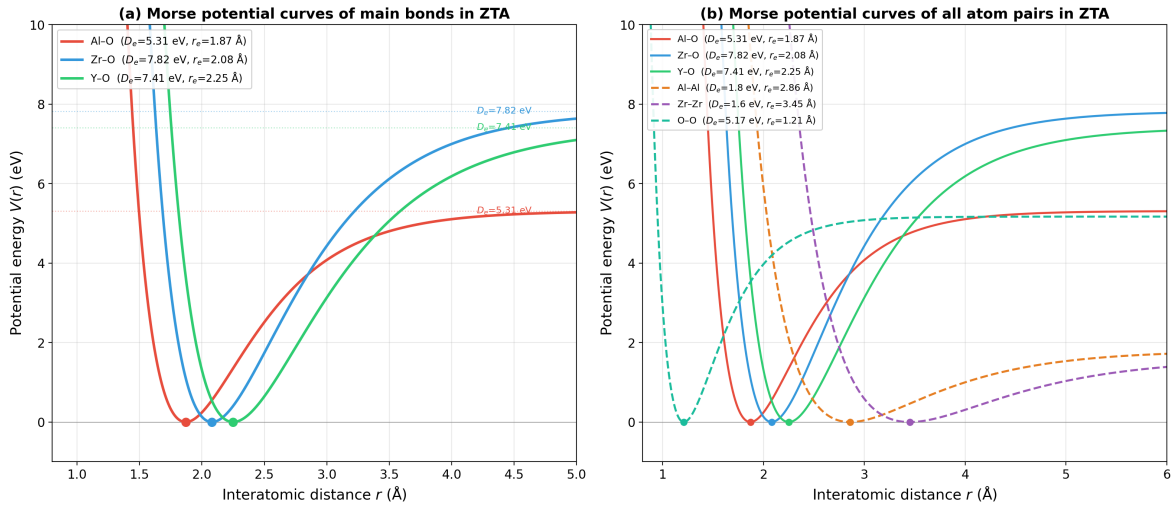


Figure 2-1-1. Morse potential curves for atomic pairs in ZTA ceramic, (a) Al–O, Zr–O, Y–O primary bonds;(b) all atomic pairs. This figure is a plot calculated by Python script.

The analysis of the radial probability density distribution and overlap of electron clouds (Figure 2-1-2) further reveals, from a quantum mechanical perspective, the competitive relationship between covalent overlap and ionic attraction. The overlap between the Al 3p electron cloud and the O 2p electron cloud is the most significant (overlap integral S ≈ 0.72), consistent with the 43.3% covalency of the Al–O bond; The Zr 4d electron cloud distribution is more diffuse, showing less overlap with the O 2p cloud (S ≈ 0.55); however, the Zr–O bond still exhibits the highest bond energy (7.82 eV) due to long-range Coulombic forces resulting from its 67.1% ionic character; Y exhibits the smallest overlap of its 4d electron cloud (S ≈ 0.48), corresponding to the highest ionic character of 70.8% in the Y–O bond. The non-monotonic relationship between the

overlap integral and bond energy indicates that ionic character makes a significant contribution to the bond energies of the Zr–O and Y–O bonds.

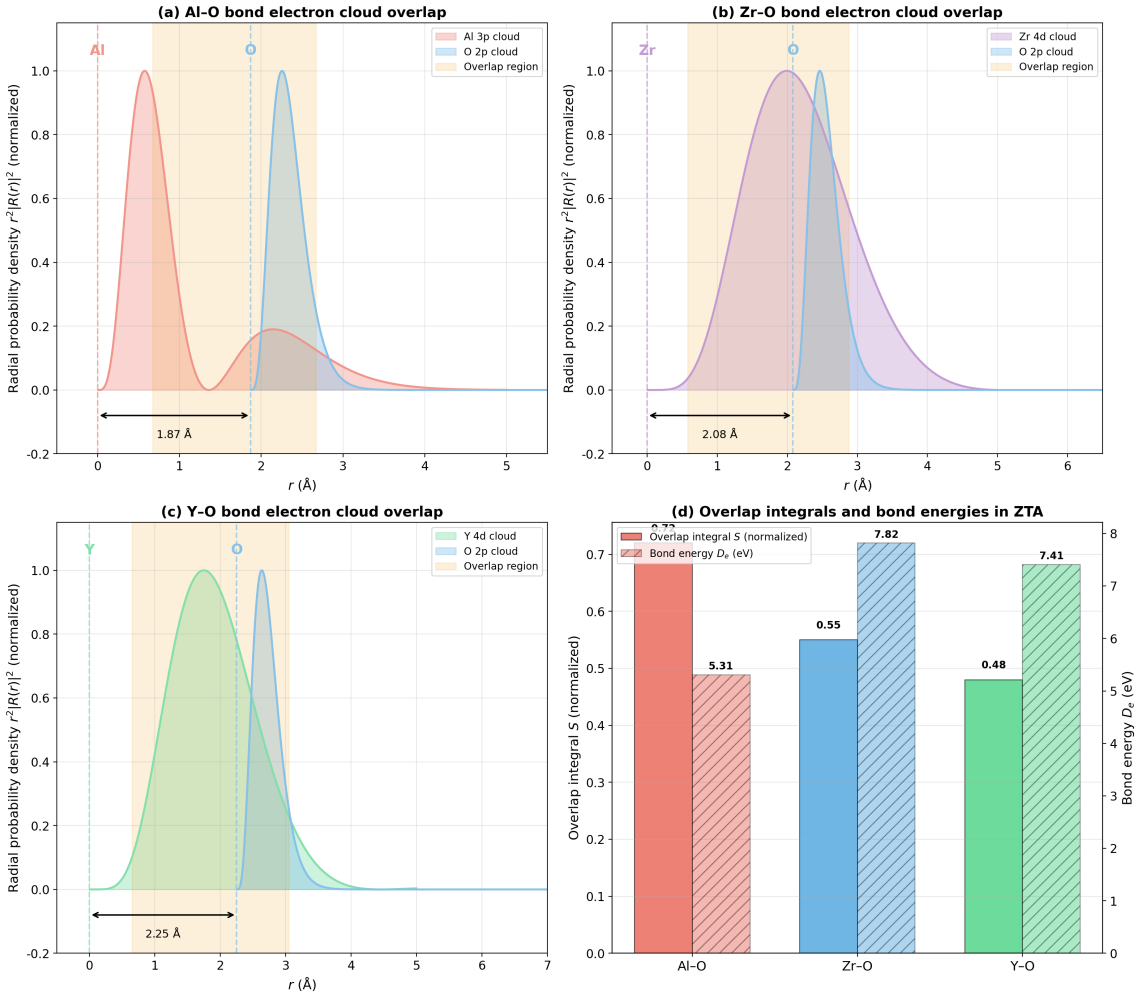


Figure 2-1-2. Electron cloud radial distribution and overlap analysis for ZTA ceramic bonds. (a)–(c): Al–O, Zr–O, Y–O overlap; (d) overlap integral vs. bond energy. This figure is calculated by a Python script.

2.2 Crystal Structure

The key crystallographic parameters of the various ZTA phases are shown in Table 2-2-1.

Table 2-2-1. Crystallographic parameters of the phases comprising ZTA

Phase	Crystal system	Group in space	Lattice parameters	Key Wyckoff Levels	Zr-O/Al-O bond length (Å)	Density (g/cm³)

α-Al_2O_3	Trigonal	R-3c（No. 167）	a = b = 4.758 Å, c = 12.991 Å [9]	Al: 6c (0, 0, z); O: 18e (x, 0, ¼)	1.87 [9]	3.99
t-ZrO_2	Tetrahedral	$P4_2/nmc$（No. 137）	a = b ≈ 3.60 Å, c ≈ 5.225 Å [10]	Zr: 2a (0, 0, 0); O: 4d (0, ½, z)	2.08 [10]	~6.10
m-ZrO_2	Monoclinic	$P2_1/c$（No. 14）	a ≈ 5.15 Å, b ≈ 5.21 Å, c ≈ 5.32 Å, β ≈ 99.2° [11]	Zr: 4e (x, y, z); O(1): 4e; O(2): 4e	2.07 [11]	~5.83

2.2.1 α-Al_2O_3 matrix phase

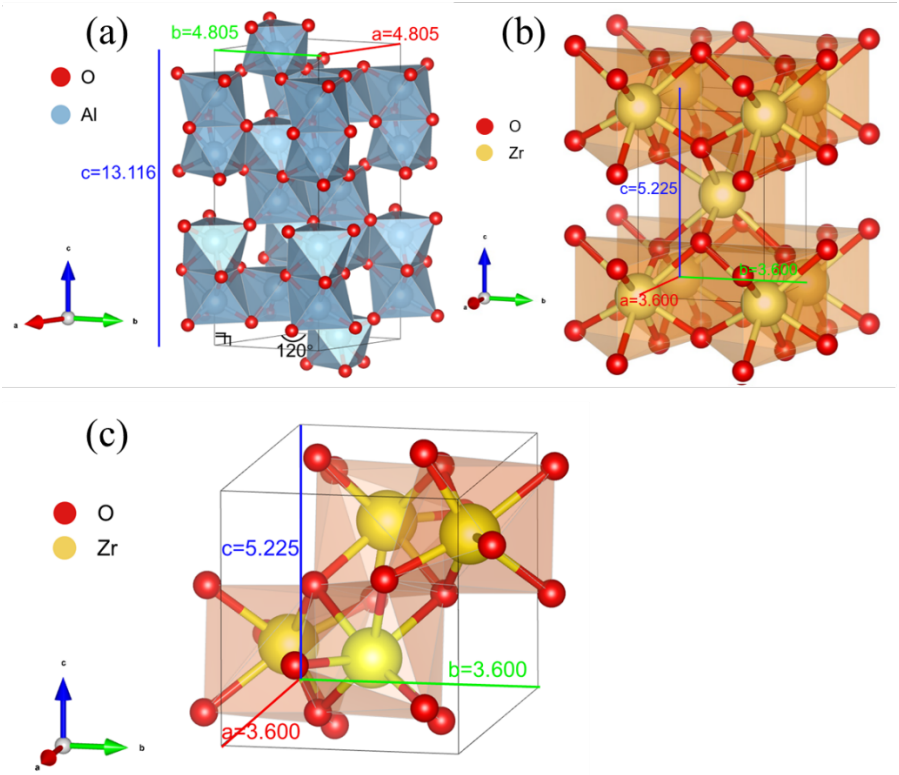


Figure 2-2-1. Crystal structure of (a)α-alumina,(b)t-zirconia and (c)m-zirconia. (This image uses crystal data provided by the Materials Project and was drawn and labeled by the author.) [21,23,29]

The α-Al_2O_3 corundum structure features oxygen atoms arranged in an hcp lattice, with Al^{3+} ions occupying 2/3 of the octahedral voids. AlO_6 octahedra are connected via shared edges and faces to form a dense three-dimensional network, with an atomic packing density of 90.7%, imparting high hardness and excellent wear resistance.

The t-ZrO_2 structure is a tetragonal fluorite derivative in which Zr^{4+} occupies the 2a site and is coordinated by 8 O^{2-} atoms in a distorted cubic arrangement. The displacement

of O^{2-} along the c-axis (z ≈ 0.20) induces tetragonal distortion (c/a ≈ 1.45), which serves as the structural driving force for the t→m phase transition.

The m-ZrO_2 monoclinic structure exhibits reduced symmetry, with the coordination number of Zr^{4+} decreasing from 8 to 7. The β angle is approximately 99.2°, and the unit cell volume expands by about 4.4% compared to t-ZrO_2.

In ZTA, t-ZrO_2 achieves room-temperature retention through a triple synergistic effect: Y_2O_3 doping introduces oxygen vacancies that stabilize the tetragonal structure; grain size is controlled below the critical size (~0.5 μm) to enhance metastability; and the Al_2O_3 matrix exerts compressive stress to suppress spontaneous phase transformation.

2.3 Polymorphs and Phase Stability

The mechanical properties and service stability of ZTA highly depend on the precise control of the stability of Al_2O_3 and ZrO_2 phases.

2.3.1 Polymorphs of Al_2O_3

α-Al_2O_3 is the only thermodynamically stable phase in the alumina system; the transition phases γ, δ, and θ are all metastable phases that gradually transform into the α phase upon heating, accompanied by volume contraction. [13] Medical-grade ZTA must be produced using high-purity α-Al_2O_3 raw materials and ensure that the transition phase is fully transformed.

2.3.2 ZrO_2 Polymorphs and Stabilization Strategies

In equilibrium state, pure ZrO_2 will exhibit three polymorphs as the temperature changes, and their phase transition sequence can be represented as:

$$m\text{-}ZrO_2 \underset{1170℃}{\Longleftrightarrow} t\text{-}ZrO_2 \underset{2370℃}{\Longleftrightarrow} c\text{-}ZrO_2 \xrightarrow[2715℃]{} l$$

The t-m phase transition at 1170°C is of the martensitic type and is accompanied by a 3–5% volume expansion; this causes cracking in single-phase zirconia ceramics, whereas ZTA transforms this into a toughening mechanism through metastabilization.

Selection of t-ZrO_2 as the toughening phase: m-ZrO_2 lacks the ability to toughen through phase transformation; c-ZrO_2 lacks the ability to undergo phase transformation; only partially stable t-ZrO_2 can remain metastable under zero stress and undergo controlled phase transformation under stress, thereby achieving a balance between toughness and stability.

Regulation is required to maintain metastable t-ZrO_2: Y_2O_3 content of 2.5–3.5 mol%; ZrO_2 particle size controlled below approximately 0.5 μm; elastic confinement by the Al_2O_3 matrix increases the phase transformation stress threshold [14].

In ZTA, the Al_2O_3 matrix confinement, low ZrO_2 connectivity, and fine grain size significantly suppress low-temperature hydrothermal decomposition (LTD); Chevalier et al. confirmed that the monoclinic phase content remained below 5% even after ageing.[34]

The design of metastabilized t-ZrO_2 through Y_2O_3 doping and microstructural control provides a scientific basis for the comprehensive performance of ZTA.

Task 3	Phase diagrams and target phase (20 marks)	
Length	Indicative length 1000-1500 words	Xiangjun Zheng

3.1 Representative Phase Diagrams

As shown in Task 1, the composition of the target material ZTA in this study ranges from approximately 80–90 wt% Al_2O_3 and 10–20 wt% ZrO_2. Additionally, 3 mol% Y_2O_3 is typically added industrially to suppress the premature transformation of T-ZrO_2 at high temperatures, allowing it to remain stable as a toughening phase at room temperature. [35]

This can be analyzed by referring to Figures 1 and 2. In the Al_2O_3–ZrO_2 binary system, when the zirconia content is below 37%, a simple eutectic reaction occurs: at temperatures above 1860°C, the system consists of a liquid phase coexisting with corundum (Al_2O_3); At 1860°C, an eutectic reaction occurs: liquid phase → corundum (Al_2O_3) + tetragonal ZrO_2 (t-ZrO_2); upon cooling to 1150–1170°C, t-ZrO_2 transforms into monoclinic ZrO_2 (m-ZrO_2); below 1150°C, the system exists stably as a two-phase mixture of corundum and m-ZrO_2.

In the ZrO_2–Y_2O_3 binary system, at room temperature, a two-phase mixture consists of a monoclinic zirconia solid solution (Mss) with a low yttrium content and a cubic zirconia solid solution (Css) with a high yttrium content; the tetragonal zirconia solid solution (Tss) shown in the phase diagram is stable only above 1114 °C.

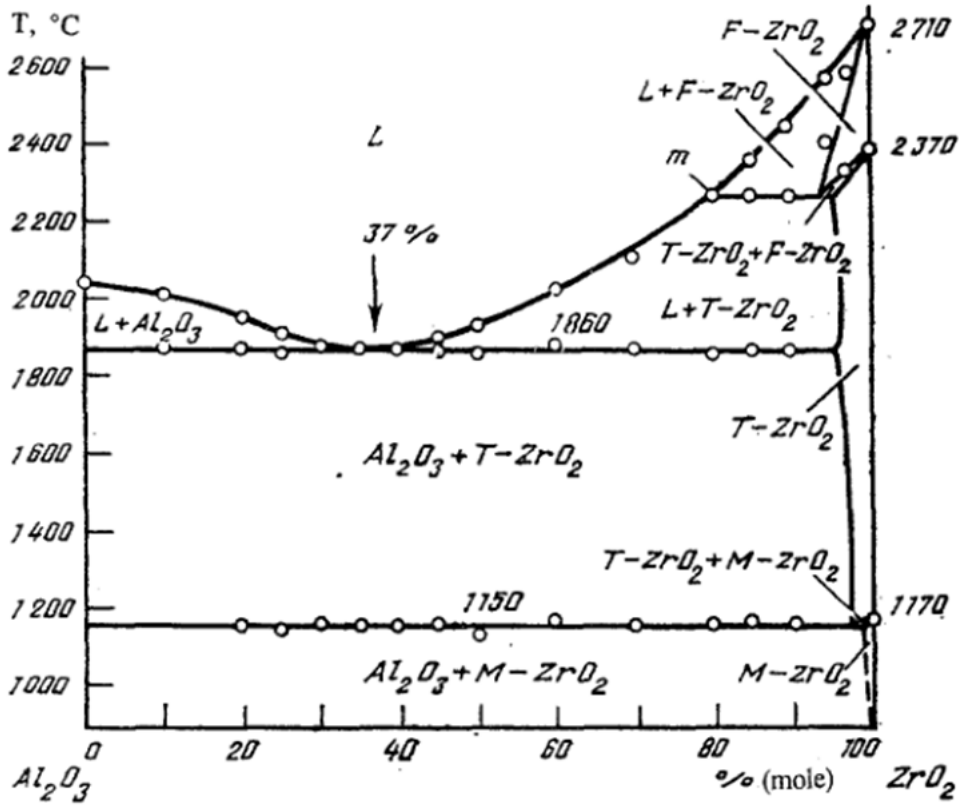


Figure 3-1-1. Phase diagram of the Al_2O_3-ZrO_2 binary system [36]

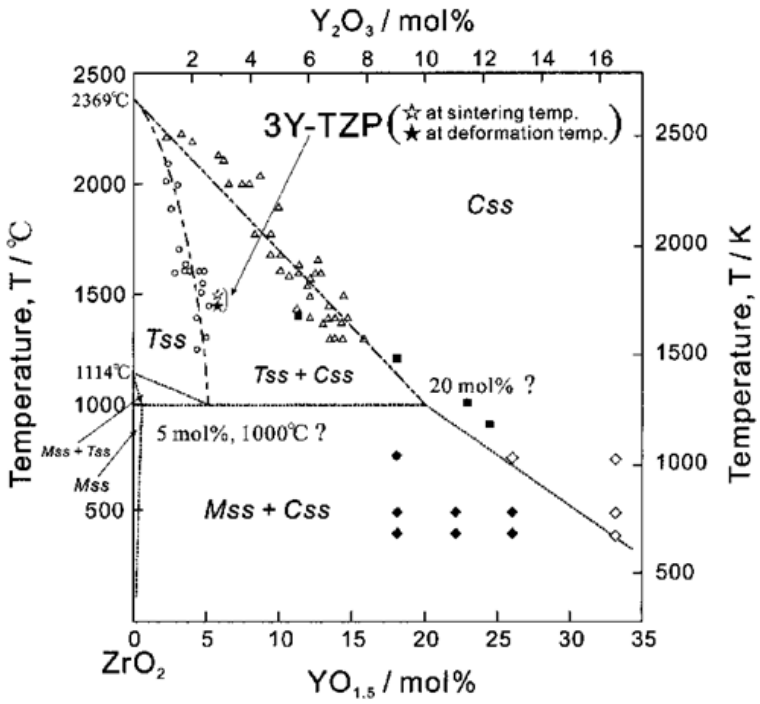


Figure 3-1-2. Phase diagram of the ZrO_2-Y_2O_3 binary system (low yttria range)[37]

As can be seen from the phase diagram, no metastable tetragonal zirconia (t-ZrO_2) particles exist at room temperature under equilibrium conditions. This contradicts the findings of Task 1 and does not align with actual industrial production. A detailed analysis of this issue will be presented in Section 3.2.2.

3.2 Phase Transformations

3.2.1 Equilibrium Phase Transition Sequence

Based on the ternary phase diagram of Al_2O_3–ZrO_2–Y_2O_3 (Figure 3), typical industrial compositions of ZTA for artificial joints were selected (80 wt% Al_2O_3, 19 wt% ZrO_2, 1 wt% Y_2O_3). Under isothermal cooling conditions, Table 1 was compiled based on Figures 1 and 2 and relevant data to illustrate the phase transformation process.

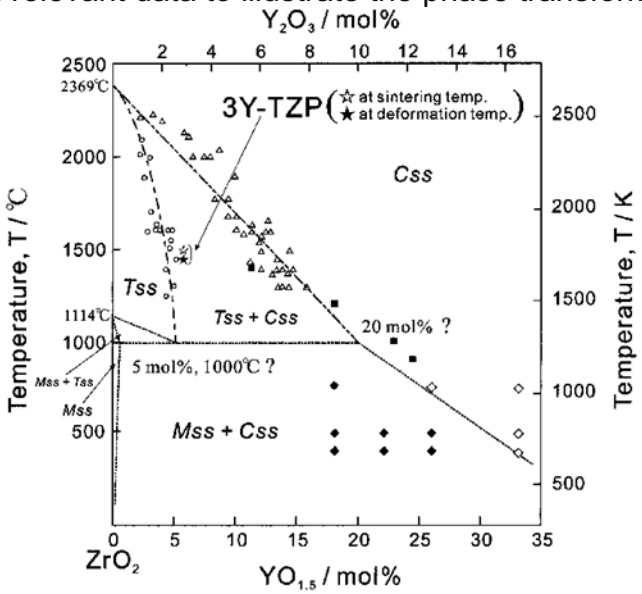


Figure 3-2-1. Projection of the liquidus surface in the Al_2O_3-ZrO_2-Y_2O_3 ternary system [38]

Table 3-2-1. Sequence of equilibrium cooling phase transitions in the 80 wt% Al_2O_3–19 wt% ZrO_2–1 wt% Y_2O_3 system

Temperature range	当前相
T > ~1960°C	L
1960°C → ~1860°C	L + α-Al_2O_3
~1860°C	α-Al_2O_3 + T-ZrO_2(ss,Y_2O_3)
1860°C → ~1170°C	α-Al_2O_3 + T-ZrO_2(ss,Y_2O_3)
~1170°C → ~1150°C	α-Al_2O_3 + M-ZrO_2(ss,Y_2O_3)
T < 1150°C	α-Al_2O_3 + M-ZrO_2(ss,Y_2O_3)

At high temperatures, this system exists as a single-phase liquid phase; upon cooling to approximately 1960°C, primary α-Al_2O_3 precipitates; as the temperature continues to drop to 1860°C, a binary eutectic reaction occurs, the liquid phase completely transforms into tetragonal ZrO_2 solid-solved with α-Al_2O_3 and Y_2O_3. When the temperature drops to 1170–1150 °C, the tetragonal solid solution undergoes a martensitic phase transformation to monoclinic ZrO_2 solid-solved with Y_2O_3. Below 1150 °C, the microstructure stabilizes with no further phase transformations.

3.2.2 Limitations of Equilibrium Thermodynamics

The difference between the two lies in the fact that the phase diagram of thermodynamic equilibrium describes phase transition behavior only under conditions of thermodynamic equilibrium, assuming that the system has sufficient time to complete atomic diffusion and structural rearrangement. In contrast, industrial sintering employs a non-equilibrium process that overcomes the limitations of thermodynamic equilibrium through kinetic control. Combined with the stabilizing effect of ionic doping in Y_2O_3, this approach enables the retention of the metastable tetragonal phase at room temperature.The following text will incorporate the revised XRD data from Wang Danning's team in Tables 3-2-2 and 3-2-3 (Tables 1 and 3 in the original paper).

Table 3-2-2. Rietveld-refined crystal structure parameters for pure ZrO_2 and doped, stabilized ZrO_2 [39]

Crystal phase	Space group	Lattice constant / nm	Atomic positions and coordinates						
			Ion	Crystal Site	x	y	z	B/nm2	N
ZrO_2	$P2_1/c$	a=0.5146(7)	Zr^{4+}	4e	0.2728(2)	0.0337(2)	0.2088(4)	0.0078	4.0
		b=0.5205(1)							
		c=0.5313(6)	O_1^{2-}	4e	0.0773(9)	0.3133(8)	0.3046(3)	0.0036	4.0
Monoclinic	Z=4	β=99.1(8)°							
		V=0.1405 nm³	O_{II}^{2-}	4e	0.4619(1)	0.7882(5)	0.4362(3)	0.0305	4.0
CeO_2-stabilized ZrO_2	$P4_2/nmc$	a=0.3626(5)	Zr^{4+}	2b	0.7500	0.2500	0.2500	0.0028	1.76
		c=0.5226(3)	Ce^{4+}	2b	0.7500	0.2500	0.2500	0.0045	0.24
Tetrahedral	Z=2	V=0.0687 nm³	O^{2-}	4d	0.2500	0.2500	0.0482(6)	0.0097	4.0
Y_2O_3-stabilized ZrO_2	$P4_2/nmc$	a=0.3602(8)	Zr^{4+}	2b	0.7500	0.2500	0.2500	0.0030	1.88
		c=0.5179(1)	Y^{3+}	2b	0.7500	0.2500	0.2500	0.0006	0.12
Tetrahedral	Z=2	V=0.0672 nm³	O^{2-}	4d	0.2500	0.2500	0.0493(8)	0.0092	3.76
Y_2O_3-stabilized ZrO_2	Fm3m	a=0.5135(9)	Zr^{4+}	4a	0.0000	0.0000	0.0000	0.0072	4.0
cubic	Z=4	V=0.1354 nm³	O^{2-}	8c	0.2500	0.2500	0.2500	0.0157	8.0

Table 3-2-3. Ionic Coordination and Bond Parameters of Y_2O_3-Stabilized Tetragonal ZrO_2 [39]

Y_2O_3-stabilized ZrO_2			
ion	Coordination number	Key length /nm	
(Zr^{4+}, Y^{4+})	8	(Zr,Y)-4O	0.2080
		4O	0.2377
O^{2-}	10	O-2(Zr,Y)	0.2080
		2(Zr,Y)	0.2377
		2O	0.2590
		4O	0.2598
O-Zr-O Key angle			
135.92°			

Based on the refined crystal structure data, the mechanism stems from the synergistic interaction between non-equilibrium kinetic effects and ionic doping stabilization: during high-temperature sintering (>1114 °C), the system resides in the single-phase region of the tetragonal phase; rapid cooling prevents the high-temperature tetragonal phase from undergoing an equilibrium phase transition, causing it to be “frozen” at room temperature to form a metastable tetragonal phase; simultaneously, the doping effect of Y^{3+} non-equivalently substituting for Zr^{4+} suppresses structural destabilization of the tetragonal phase by regulating the distribution of lattice defects and coordination states, thereby further stabilizing the metastable phase.

Refined data provide direct structural evidence for this cooperative stabilization mechanism:

1. Crystal phase and lattice parameter data show that the lattice constants of the tetragonal crystal phase of ZrO_2 stabilized by Y_2O_3 (space group $P4_2/nmc$) are a = 0.3602(8) nm and c = 0.5179(1) nm, directly confirming that the stable crystal phase of this system at room temperature is tetragonal;

2. In the atomic occupancy data, the actual occupancy of O^{2-} at the 4d site is 3.76, which is lower than the theoretical occupancy of 4.0. This discrepancy directly demonstrates that the substitution of Y^{3+} for Zr^{4+} creates oxygen vacancies to balance the charge. Such defects optimize the charge distribution in the tetragonal lattice and reduce the driving force for structural distortion;

3. Bond length and coordination data show that after doping, (Zr^{4+}, Y^{3+}) forms two distinct coordination bond lengths with O^{2-} (0.2080 nm and 0.2377 nm). The

introduction of Y^{3+} alleviates the lattice stress caused by differences in ionic radii by forming long and short coordination bonds, thereby stabilizing the coordination environment of the tetragonal phase.

Based on the refined data presented above, it can be confirmed that in industrial and practical applications, the stable existence of the tetragonal phase in low-concentration Y_2O_3-doped ZrO_2 at room temperature can be reliably achieved through a synergistic process involving high-temperature sintering to obtain a high-temperature tetragonal phase structure, rapid cooling to suppress equilibrium phase transformations, and low-concentration Y_2O_3 doping to regulate lattice defects and coordination states.

3.3 Interpretation and application alignment

In industrial production, the sintering process is one of the key factors determining the mechanical properties of ZTA ceramics.

The pressureless two-step sintering process—a widely used traditional sintering technique for ceramic materials—can be regarded as a typical example of achieving the desired microstructure through the control of non-equilibrium kinetics.

In this process, the green body is first heated at a rate of 10°C/min to 1550°C and held at that temperature for 3 minutes, allowing the body to undergo rapid initial densification at high temperature, achieving a density of 75%–90%; It is then rapidly cooled to 1300°C and held at that temperature for 10 hours. At this lower temperature, final densification is achieved through lattice diffusion, while simultaneously suppressing grain coalescence and abnormal grain growth at the kinetic level, resulting in a uniform submicron grain structure. [40]

The design of these process parameters is entirely based on the thermodynamic boundaries defined by the phase diagram: the eutectic point at 1860°C in the Al_2O_3-ZrO_2 binary phase diagram defines the upper temperature limit, ensuring that no excessive liquid phase forms in the high-temperature range at 1550°C, thereby preventing grain coarsening, green body deformation, and compositional segregation; The 1114°C eutectoid point of the ZrO_2-Y_2O_3 binary phase diagram, together with the 1150°C t→m phase transition point of the Al_2O_3-ZrO_2 binary phase diagram, jointly define the lower temperature limit. This ensures that the 1300°C holding stage remains within the thermodynamically stable range of tetragonal ZrO_2, laying the foundation for the retention of metastable phases during subsequent cooling.

Given the phase transformation characteristics of ZrO_2, the differences in crystal structure among its various polymorphs form the core basis for the t→m phase transformation and the phase-transition toughening effect, as shown in the figure:

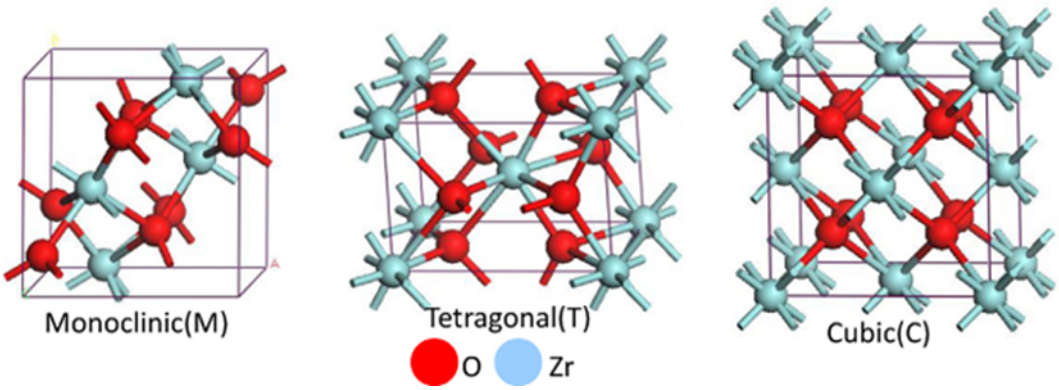


Figure 3-3-1. Schematic diagrams of the monoclinic (M), tetragonal (T), and cubic (C) crystal structures of ZrO_2 [41].

The figure clearly illustrates the three typical crystal structures of ZrO_2: the monoclinic phase (M) features a distorted heptacoordinate structure, the tetragonal phase (T) features a distorted octacoordinate structure, and the cubic phase (C) features a regular fluorite-type octacoordinate structure. During the displacement-type martensitic phase transformation, the coordination environment of Zr^{4+} changes from the distorted octahedral coordination of the tetragonal phase (T) to the distorted heptagonal coordination of the monoclinic phase (M), accompanied by a volume expansion of approximately 3%–5%—this is both the core mechanism of toughening during the ZrO_2 phase transformation and the root cause of the room-temperature brittleness of undoped ZrO_2. The rapid cooling process during two-step sintering kinetically inhibits the eutectic decomposition of Tss→Mss+Css, “freezing” the high-temperature-stable tetragonal phase at room temperature, while the mechanical confinement of the Al_2O_3 matrix further suppresses the phase transformation and growth of ZrO_2 grains. This ultimately yields an ideal microstructure characterized by uniform grains, the absence of abnormal grain growth, and a high concentration of the metastable t-ZrO_2 toughening phase, fully meeting the performance requirements of medical ZTA ceramics for high density, high strength, and high toughness.

Task 4	Starting raw material, green body processing technique and production volumes (20 marks)	
Length	Indicative length 1000-1500 words	Yongqi Xu, Xuanming Che

4.1 Powder Preparation

The preparation of ZTA requires high-purity α-Al_2O_3 and 3Y-TZP (ZrO_2–Y_2O_3) powders. Al_2O_3 is typically obtained from bauxite through the Bayer process and calcination. [42] ZrO_2 is derived from the refining of zircon sand and requires Y_2O_3 to stabilize the tetragonal phase, preventing pure ZrO_2 from transforming into m-ZrO_2 at room temperature. Since Y_2O_3 is a rare earth oxide subject to significant cost and supply chain fluctuations, this report prioritizes commercially available pre-stabilized 3Y-TZP. The powder purity should be at least 99.9%.

Particle size directly affects sintering activity, phase stability, and final defects. Fine powders have higher surface curvature, and the sintering driving force can be approximated as

$$\Delta P \approx \frac{2\gamma}{r}$$

where γ is the surface energy and r is the particle radius.

Therefore, smaller particle sizes help reduce the sintering temperature and improve density. However, ZrO_2 particles are not necessarily better the finer they are: if they are too small, t-ZrO_2 becomes overly stable, making it difficult for the t→m phase transformation to occur at crack tips; if they are too large, they may spontaneously transform into m-ZrO_2, leading to microcracks and surface roughening. Therefore, ZrO_2 toughening should be controlled within a critical window of approximately 0.2–0.6 μm. [43,44] This allows it to maintain a metastable tetragonal phase at room temperature while undergoing a controllable phase transformation under stress. The Al_2O_3 powder should also have a submicron size and a narrow particle size distribution to enhance sintering activity and minimize abnormal grain growth.

Particle size distribution, morphology, and agglomeration state are equally critical. A broad particle size distribution can increase bulk density, but coarse particles or hard agglomerates can lead to microstructural heterogeneity; particles with high aspect ratios or irregular shapes may also reduce flowability, affecting the density of the resulting green body. For brittle ceramics, the ultimate strength is often governed by the largest defect, which can be expressed using the Griffith criterion as

$$\sigma_f \approx \frac{K_{IC}}{Y\sqrt{\pi c}}$$

where c represents the effective defect size.

Consequently, hard agglomerates, chemical inhomogeneities, or localized ZrO_2 enrichment zones in the powder may transform into pores, abnormal grains, residual stress concentration zones, or localized m-ZrO_2 after sintering, thereby reducing flexural strength, Weibull reliability, and resistance to slow cracking.

The table below lists the suppliers of the four powders and the specifications of the products they offer.

Table 4-1-1. The suppliers of the four powders and the specifications of the products they offer

Power	Supplier	Purity	Particle size	Price
α-Al_2O_3	MSE Supplies	>99.99%	d50 ≈ 500 nm	$278/kg
ZrO_2	MSE Supplies	>99.9%	Average size 20–40 nm	$285.9/kg
Y_2O_3	MSE Supplies	>9.999%	Average size 5–8 μm	$399.95/100g
3Y-TZP	Saint-Gobain ZirPro	>99.95%	d50 ≈ 0.2–0.3 μm	$177/kg

In the subsequent preparation of ZTA composite powders, this report utilized high-purity α-Al_2O_3 powder and commercial 3Y-TZP powder, which were deagglomerated and uniformly mixed via wet ball milling. Wet dispersion reduces dust pollution, improves the uniformity of the two-phase mixture, and minimizes soft agglomeration. If intermediate storage is required, the mixed powder can be dried, sieved, and stored in sealed containers to prevent moisture absorption, contamination, and re-agglomeration. If proceeding directly to green body preparation, the slurry solids content should be adjusted before proceeding to the subsequent spray granulation step, as outlined below:

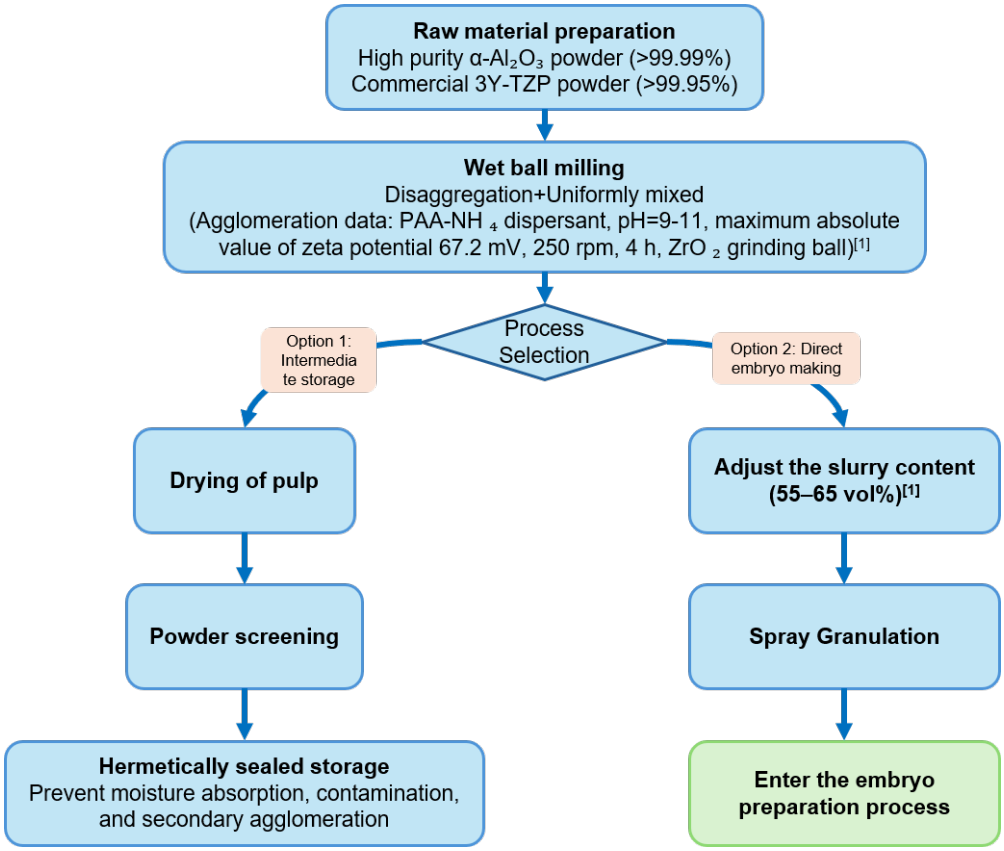


Figure 4-1-1. Flowchart of the ZTA composite powder preparation process [45]

This process better meets the requirements of hip joint ZTA for high density, low defects, and phase-transition toughening than simple dry mixing alone; it is also suitable for subsequent pre-compression and cold isostatic pressing to produce high-quality green bodies.

4.2 Green Body Preparation

This report selected the “spray granulation—low-pressure pre-compaction—cold isostatic pressing (CIP)” process as the route for preparing green bodies for ZTA ceramic hip joint heads. The core objective of this route is to transform the Al_2O_3–3Y-TZP slurry or granulated powder, which has already been dispersed as described in Section 4.1, into a green body with high bulk density, low residual air content, low density gradient, and uniform distribution of the two phases, thereby meeting the requirements for batch consistency, dimensional accuracy, and post-sintering reliability of hip joint heads.

The choice of forming method depends primarily on production efficiency, density uniformity, and the risk of defects during debinding. Unidirectional dry pressing is highly efficient but prone to axial density gradients; slip casting is suitable for complex shapes but involves long cycle times and is prone to settling or cracking; gel casting and ceramic injection molding offer better uniformity and the ability to produce complex shapes, but they have high organic content and carry a greater risk of debinding porosity. Therefore, spray granulation combined with low-pressure pre-pressing and CIP is more suitable for ZTA hip joint ceramics, as it balances medium- to high-volume production, near-net-shape preparation, green body uniformity, and reliability in subsequent sintering.[46]

The green body preparation process is as follows:

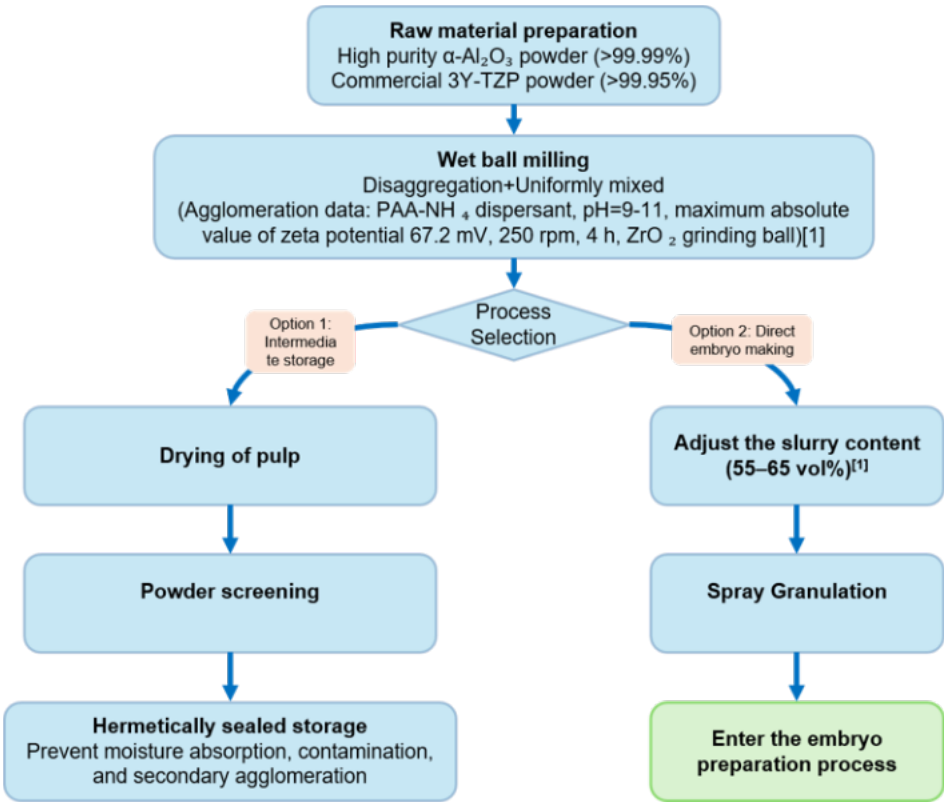


Figure 4-2-1 Process Flow for the Preparation of ZTA Composite Powder Green Bodies.[45,47]

During the slurry stabilization stage, it is necessary to control the pH, the amount of dispersant added, the solids content, and the concentration of residual ions. While Al_2O_3 and ZrO_2 are not easily subject to further oxidation by water, an inappropriate pH or an excess of dispersant can lead to flocculation, re-agglomeration, or organic

residues, thereby reducing sintering activity. Spray granulation can improve powder flowability and uniformity of mold filling; however, if solid content, viscosity, or drying rate are not properly controlled, hollow particles or hard agglomerates may form. Therefore, screening is required after granulation, and residual air between particles must be reduced through tamping or vacuum degassing prior to pre-compaction.

The purpose of low-pressure pre-compaction is not to achieve final densification, but to form a transportable near-net-shape green body and provide a stable shape for CIP. This is followed by cold isostatic pressing, which applies pressure uniformly to the green body from all directions, reducing the density anisotropy caused by unidirectional pressing. This process requires simultaneous control across three scales: the submicron scale for uniform ZrO_2 dispersion; the tens to hundreds of microns scale for granule morphology and hollow particles; and the centimeter scale for overall density uniformity.[48]

4.3 Firing and Heat Treatment

The sintering objective for ZTA hip joint ceramics is to achieve a microstructure that is nearly fully dense, with low residual porosity, fine grains, and t-ZrO_2 stabilization. The core trade-off in heat treatment is that higher temperatures and longer holding times promote diffusion densification but also facilitate the growth of Al_2O_3 and ZrO_2 grains; if ZrO_2 grains exceed the appropriate stability window, spontaneous t→m phase transformation may occur during cooling or in hydrothermal environments, leading to volume expansion, microcracks, and surface roughening. Therefore, this report adopts the following process route: degreasing → two-step unpressurized sintering → HIP post-treatment → diamond grinding and precision polishing.

Unpressurized sintering is low-cost and offers good dimensional adaptability, making it suitable for the mass production of near-net-shape ZTA green bodies; although hot pressing and SPS can rapidly densify and refine grains, they are subject to significant mold and dimensional constraints; HIP is not suitable for the direct treatment of green bodies containing organic matter or open pores, but it is suitable as a post-treatment after unpressurized sintering to eliminate residual closed pores and improve reliability.[46,49]

The specific heat treatment regimen can be designed as follows: First, heat the sample in air at a rate of 0.5–1 °C/min to approximately 500–600 °C and hold for 1–2 hours to remove PVA, dispersants, and other organic compounds. This is followed by a two-step unpressurized sintering process: first, heat to approximately 1450 °C and hold briefly to promote rapid densification; then the temperature is lowered to 1350–1400 °C and held for 6–12 h to allow densification to continue while suppressing grain coarsening. Finally, HIP can be performed in an Ar atmosphere, for example at 1350–1450 °C, 100–150 MPa, and for 1–2 h, to further reduce the residual closed-pore content.

This sintering curve should be consistent with the phase stability of ZrO_2 described in Task 3. The toughening of ZTA relies on the stress-induced t→m phase transformation of metastable t-ZrO_2 at the crack tip; therefore, heat treatment should not merely aim for the highest sintering temperature. If the temperature is too low, residual voids may serve as crack initiation sites; if the temperature is too high or the holding time is too

long, grain coarsening will weaken the controllable phase transformation toughening and increase the risk of aging.

To estimate the grain size after sintering, an empirical grain growth model can be used:

$$G^n - G_0^n = kt$$

where G is the grain size after sintering, G_0 is the initial grain size, n is the grain growth exponent, t is the holding time, and k is the temperature-dependent rate constant, and:

$$k = k_0 \exp(-Q/RT)$$

This report uses literature parameters for Al_2O_3 grain growth in ZTA for estimation, taking n = 3.97 and Q = 429.19 kJ mol^{-1}, and assuming that the initial grain size of the Al_2O_3–3Y-TZP composite green body after wet ball milling is G_0 = 50 nm.

The calculation results indicate that grain growth is highly temperature-sensitive. A prolonged hold at 1350 °C can still maintain a fine-grained structure, whereas a prolonged hold at 1450 °C significantly accelerates grain coarsening. Therefore, the final process should not involve a prolonged hold at temperatures above 1450 °C; instead, a two-step sintering route of “short-term densification at 1450 °C followed by a relatively long hold at 1350–1400 °C” should be adopted.

The high-temperature furnace environment has little effect on the composition of the main phase, as Al_2O_3, ZrO_2, and Y_2O_3 do not readily sublimate or volatilize in air or an Ar atmosphere at 1350–1450 °C. It is even more critical to control impurities such as Na_2O, B_2O_3, SiO_2, and residual Cl^- and NO_3^-, as these may form low-melting-point grain boundary phases and reduce grain boundary strength. Therefore, the control of powder purity, ball mill media wear, residual dispersants, and storage contamination in Task 4.1 will directly affect sintering quality.

After sintering, the components must undergo cutting, diamond grinding, and precision polishing to achieve the low-roughness surface required for hip joint bearings. Polishing reduces friction and the generation of wear particles, but grinding may introduce surface microcracks, residual stresses, and localized t→m phase transformations, which in turn reduce strength, fatigue life, and wear resistance. Therefore, Task 5 should focus on characterizing the phase composition, grain size, relative density, surface roughness, grinding damage, and wear behavior.

Task 5	Analytical characterizations, testing and device quality assurance (20 marks)	
Length	Indicative length 1000-1500 words	Xuanming Che

5.1 Analytical Characterisations

The ideal ZTA sintered ceramic consists primarily of α-Al_2O_3 and Y-stabilized t-ZrO_2, without m-ZrO_2 or impurities. According to the requirements of Task 4, the ideal size of ZrO_2 particles should be 0.2–0.6 micrometers, and they should be uniformly distributed within the Al_2O_3 matrix; Al_2O_3 grains should also remain fine and uniform. The sintered body should be nearly fully dense and free of visible defects. As shown in the table, SEM is used to confirm the microstructure of the ZTA, the ZrO_2 particle size distribution, and the presence of pores or defects; EDS is used to verify whether the Y element is primarily enriched in the ZrO_2 phase; XRD is used to confirm whether the desired t-ZrO_2 phase is retained after sintering and to rule out the undesirable m-ZrO_2 phase.

Table 5-1-1. Summary of selected characterization techniques used for quality control of the prepared ceramic materials, listed in descending order of importance.

Techniques	**Analytical characterization**	**Morphological characterization**
SEM	Atomic number	Particle morphology and size
XRD	Crystal structure, phase composition	-
EDS	Identification of Components (Semi-quantitative)	Element Distribution
Raman	Phase Composition (Supplement to XRD)	-
Archimedes	Density	-

5.1.1 SEM

Figure 5-1-1(a) shows the microstructure of ZTA (10 vol% ZrO_2). Due to the high atomic number of Zr, the bright phase represents ZrO_2, while the dark phase represents Al_2O_3; ZrO_2 is uniformly dispersed within Al_2O_3. Y_2O_3 is not visible as distinct particles. (b) The figure shows the particle size distribution of ZTA. Dc is the critical size for spontaneous transformation after sintering, above which there is no reinforcement; D'c is defined as the critical size for transformation during crack propagation, below which there is no crack-induced transformation, and thus no reinforcement. Combining this with the critical size window determined from the literature in Task 4.1, D'c ≈ 0.2 μm can be regarded as the lower limit for crack-induced phase transformation, and Dc ≈ 0.6 μm as the upper limit for spontaneous phase transformation during cooling. Most ZrO_2 grains fall within this range. Due to the small size and narrow size distribution of ZrO_2 grains, a large number of tetragonal ZrO_2 grains can be retained in the alumina matrix, which facilitates the phase transformation toughening mechanism in these composites.

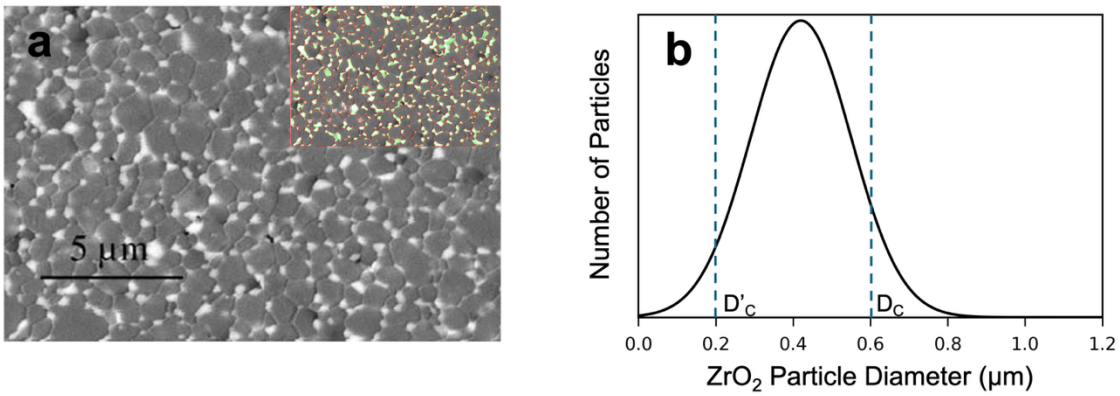


Figure 5-1-1. (a) Scanning backscatter electron microscope image of ZTA (10 vol% ZrO2). ZrO2 grains (brighter phase) are uniformly distributed in a fine-grained Al2O3 matrix (darker phase). Al2O3 D50=1.27±0.4 μm and ZrO2 2 D50=0.47±0.1 μm (Based on an image from the paper; the top-right corner shows a schematic illustrating the use of Python to count grains, by us.) (b)Grain size distribution of ZTA (10 vol% ZrO_2). The dashed lines indicate the critical size for spontaneous transformation (Dc) and the critical size for crack propagation (D'c) after sintering (redrawn from the original figure after removing interfering information, by us).[50]

5.1.2 EDS

EDS is commonly used to analyze elemental composition. Figure 5-1-2 shows the elemental distribution in the ZTA. Al is detected in the dark regions, while Zr and Y signals are detected in the bright regions and spatially overlap, indicating that Y is primarily distributed within the ZrO_2 phase. This is due to the stabilization of oxygen vacancies caused by Y_2O_3 doping, which can be described by the Kröger–Vink reaction:

$$Y_2O_3 \xrightarrow{ZrO_2} 2Y'_{Zr} + V_{\ddot{O}} + 3O_O^{\times}$$

The physical significance is as follows: when Y^{3+} replaces Zr^{4+}, oxygen vacancies $V_{\ddot{O}}$ are introduced to maintain charge compensation; these oxygen vacancies reduce the free energy of the tetragonal phase, allowing t-ZrO_2 to persist in a metastable form at room temperature. Next, XRD is required to identify the phase in which ZrO_2 exists.

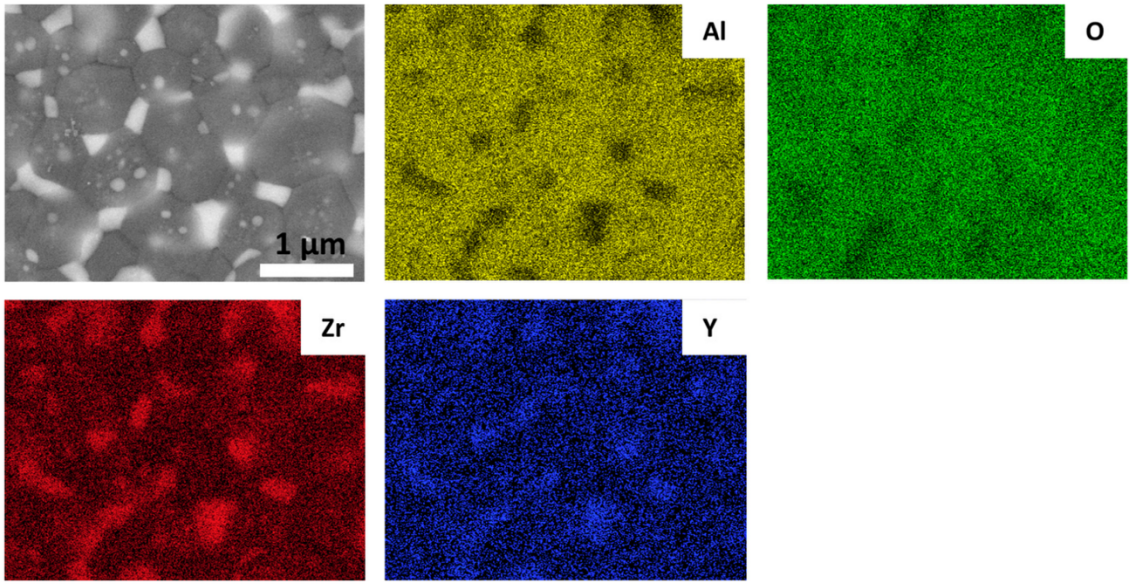


Figure 5-1-2. EDS images of Al, O, Zr, and Y in ZTA (10 vol% ZrO2)[51]

5.1.3 XRD

Figure 5-1-3 shows the XRD patterns of the green bodies and ZTA ceramics at different microwave sintering temperatures. The patterns of the green bodies indicate that the as-synthesized ZTA ceramics contain m-ZrO_2 and Al_2O_3. The characteristic (-111) and (111) peaks of m-ZrO_2 are typically located at approximately 28° and 31.5°, while the (101) peak of t-ZrO_2 is located at approximately 30°. At 1350°C, the m-ZrO_2 peaks disappear, and distinct t-ZrO_2 peaks become observable. This indicates that, as the sintering temperature increases, the phase transforms from m-ZrO_2 to t-ZrO_2, and the peak intensity of t-ZrO_2 continuously increases.

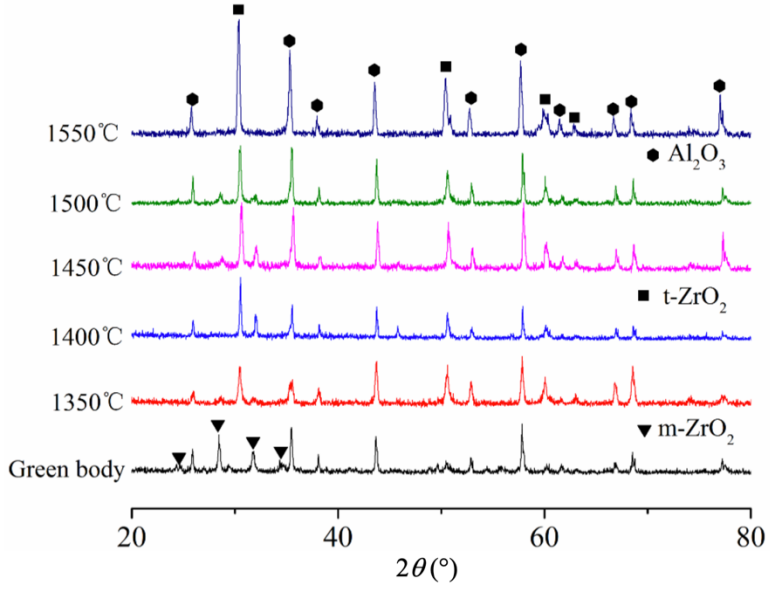


Figure 5-1-3. XRD patterns of ZTA (22 wt% ZrO_2)[52]

In addition, the reliability of the test results must be ensured through instrument calibration. XRD peak positions should be calibrated using standard reference materials such as NIST LaB_6.[53]. EDS quantitative analysis requires control of beam stability, acceleration voltage, and detector dead time[54]. SEM image analysis

requires first calibrating the pixel size using the scale bar in the image, followed by particle identification and particle size statistics using Python.[55]

Therefore, SEM confirmed the particle size and distribution of ZrO_2, EDS confirmed that the Y element was primarily incorporated into the ZrO_2 phase, and XRD confirmed the presence of the t-ZrO_2 phase after sintering.

Overall, Section 5.1 demonstrates that the crystal structure and phase stability design in Task 2, as well as the sintering process in Task 4, have resulted in the desired α-Al_2O_3 + retained t-ZrO_2 microstructure, providing a foundation for subsequent verification of mechanical properties.

5.2 Properties Testing

5.2.1 Bending Test

The flexural strength was determined using a biaxial flexural strength test, specifically the piston-on-three-ball test. Circular specimens with a diameter of 16 mm and a thickness of 1.6 mm were prepared and tested on a universal testing machine at a crosshead speed of 0.5 mm/min until fracture. [56]

First, ZTA (0, 10, 20, and 30 vol% ZrO_2) was tested. The results showed that as the 3Y-TZP content increased from 0 vol% to 30 vol%, the biaxial flexural strength of ZTA improved significantly. This increase in strength is primarily attributed to several factors: the ZrO_2 second phase inhibits Al_2O_3 grain growth, leading to grain refinement in the matrix; ZrO_2 induces crack deflection; and the stress-induced phase transformation from t-ZrO_2 to m-ZrO_2 combined with residual compressive stress collectively impede crack propagation.

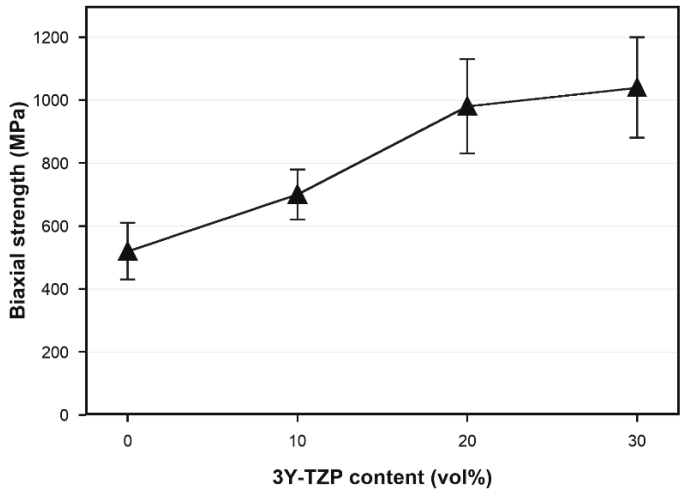


Figure 5-2-1. Effect of 3Y-TZP content on the biaxial flexural strength of sintered specimens. (Redrawn using data points extracted from the paper)

Next, 30 samples of ZTA (0, 20 vol% ZrO_2) were tested, and Weibull plots were generated. Since ceramics are brittle materials and their fracture strength is governed by the largest defect, it is essential to evaluate the dispersion of the strength data. Weibull analysis describes the strength distribution in terms of failure probability,

providing an effective means of assessing a material's reliability. The two-parameter Weibull distribution can be expressed as:

$$P_f = 1 - exp\left[-\left(\frac{\sigma}{\sigma_0}\right)^m\right]$$

P_f is the failure probability, σ is the fracture strength, σ_0 is the characteristic strength, and m is the Weibull modulus. A higher value of m indicates that the strength data is more concentrated, the distribution of material defects is more uniform, and reliability is higher.

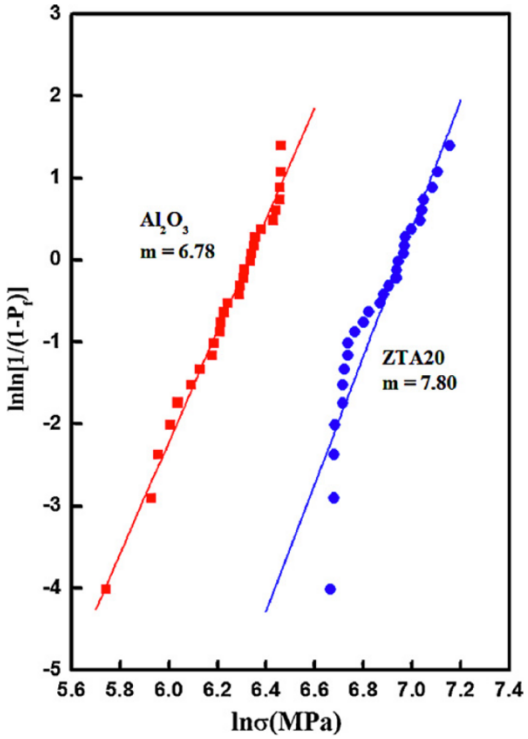


Figure 5-2-2. Weibull plot of ZTA composites. The solid line was obtained by fitting the data using the least-squares method. (Figure from the original paper)

Figure 5-2-2 shows the Weibull plots for the alumina and ZTA20 composites. The Weibull modulus of the ZTA20 sample (7.8) is higher than that of alumina (6.78), indicating a slight improvement in the reliability of the ZTA20 composite. The strength distribution of ZTA20 appears to be bimodal or exhibits a non-Weibull distribution. It has been reported that internal compressive stress can influence the strength distribution. [57] Because ZrO_2 and Al_2O_3 have different coefficients of thermal expansion, residual stresses—including radial tensile stress and hoop compressive stress—form in the composite material after sintering and cooling. This hoop compressive stress can deflect cracks and increase the crack propagation paths, thereby improving strength and fracture toughness; however, it also causes the strength distribution to deviate from simple single-Weibull behavior.

From a design perspective, the bimodal or non-uniform Weibull distribution of the ZTA20 means that judgments cannot be based solely on the mean strength. For ceramic femoral heads, safety is often determined by samples in the low-strength tail of the distribution—that is, the high-risk group corresponding to larger defects, localized agglomeration, or adverse residual stress conditions. Therefore, a

conservative design should be based on the reliability of the Weibull low-tail distribution, rather than merely comparing mean flexural strengths.

5.2.2 FEA-assisted Contact Reliability

To link laboratory bending strength and Weibull reliability analysis to real-world hip joint contact conditions, this report establishes a two-dimensional axisymmetric finite element contact model. ZTA ceramic is treated as a linearly elastic isotropic material with a Young's modulus E = 358 GPa and a Poisson's ratio v = 0.24.[58] The femoral head radius is 18 mm, the reference radial gap is 0.10 mm, and the outer surface of the cup liner is fixed. An axial load of 3000 N is applied as a distributed pressure over a small area above the femoral head. This loading method avoids the singularities associated with point loads. According to Saint-Venant's principle[59], ince the loading region is far from the contact region, the stress field in the contact region is not significantly affected by the loading details. Contact is defined as frictionless face-to-face contact.

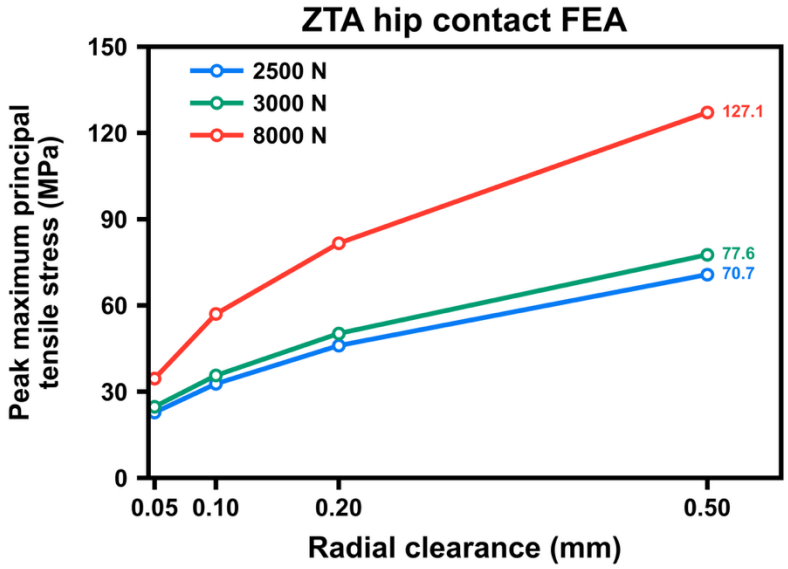


Figure 5-2-3. Effect of radial clearance on the peak maximum principal tensile stress under different loads. As the load and radial clearance increase, the peak maximum principal tensile stress rises, indicating that a larger clearance reduces the contact fit between the ceramic components and causes the load to become more concentrated.

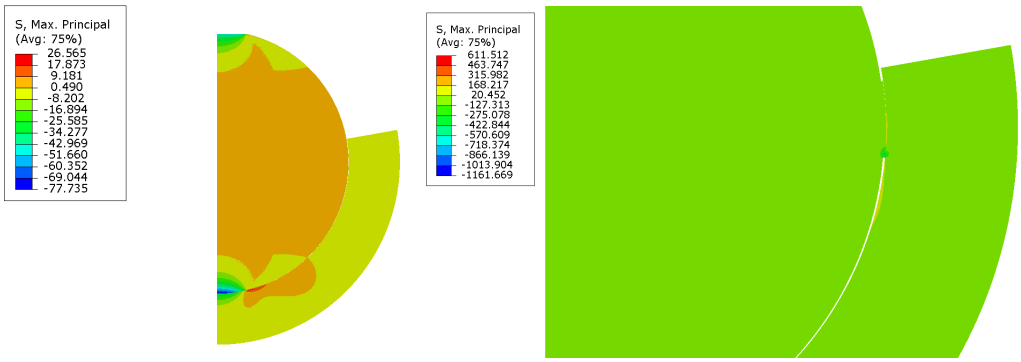


Figure 5-2-4. Maximum principal stress contour plot at 3000 N. (a) Centered reference contact condition, with a peak maximum principal tensile stress of 26.6 MPa; the stress region at the top of the figure corresponds to the load application area, not the second

contact zone. (b) Slightly separated/edge contact condition, with a peak maximum principal tensile stress of 611 MPa; high stress is concentrated at the rim of the cup.

The results of the parametric analysis show that the peak principal tensile stress increases with increasing load and radial clearance. This trend indicates that a larger radial clearance reduces the geometric fit of the ceramic-ceramic contact, causing the load to concentrate within a smaller contact area.

The peak maximum principal tensile stress obtained from the centered reference contact cloud map is 26.6 MPa, while the value corresponding to a 3000 N load and a 0.10 mm gap in the parametric curve is 35.6 MPa. Both values fall within the reasonable range of 26–36 MPa. To facilitate a direct comparison between edge contact and centered contact, this paper uniformly adopts the value of 26.6 MPa extracted from the contour plot as the centered reference value in the following discussion.

Under the same 3000 N load, the peak maximum principal tensile stress in the micro-separation/edge-contact condition reached approximately 611 MPa, which is about twenty times that of the centered reference contact. Therefore, cup-edge contact caused by micro-separation is a more critical design control condition in ZTA ceramic hip joints. Since ceramic failure is primarily governed by tensile stress and crack propagation, this paper uses only the maximum principal tensile stress as a failure-related indicator, rather than von Mises stress.

By substituting the finite element results for edge contact into the Weibull failure probability model in ZTA20, we can obtain the statistical failure risk under contact conditions:

$$P_f = 1 - \exp\left[-\left(\frac{\sigma_{\mathrm{FEA}}}{\sigma_0}\right)^m\right]$$

σ_{FEA} = 611 MPa， m = 7.8， σ_0 is the characteristic strength extracted from the Weibull plot of the flexural strength of ZTA20. Based on the point where y = 0—that is, the intersection where the failure probability is 63.2%—σ_0 is approximately 1080 MPa. Therefore:

$$P_{\mathrm{f}} = 1 - \exp\left[-\left(\frac{611}{1080}\right)^{7.8}\right] \approx 1.2\%$$

5.2.3 Fracture Toughness

Failure in ceramics primarily manifests as brittle fracture, and strength is highly dependent on pre-existing defects in the material, such as pores, scratches, agglomerates, and microcracks. Under applied tensile stress, the local stress field near the crack tip can be described by the stress intensity factor K_I. When K_I reaches a certain critical value, the crack undergoes rapid unstable propagation; this critical value is known as the fracture toughness K_{IC}. Therefore, K_{IC} characterizes a material's ability to resist rapid catastrophic fracture; the higher the K_{IC}, the larger the defect size or applied load the material can tolerate before fracturing.

The double-torsion testing method is used to evaluate the crack propagation behavior of materials. First, the tensile surface of the specimen is polished, then an initial notch is machined, and a true sharp crack is created through low-load indentation and pre-cracking steps. The testing primarily consists of two modes: the first is a load-relaxation test, used to measure crack velocities in a higher range; the second is a constant-load test, used to measure extremely low crack velocities. Using the obtained $V - K_I$ curve, the stress intensity factor corresponding to the point where the crack velocity drops to an unmeasurable level is defined as K_{I0}, and K_{IC} is obtained by extrapolating the curve into the rapid fracture region.

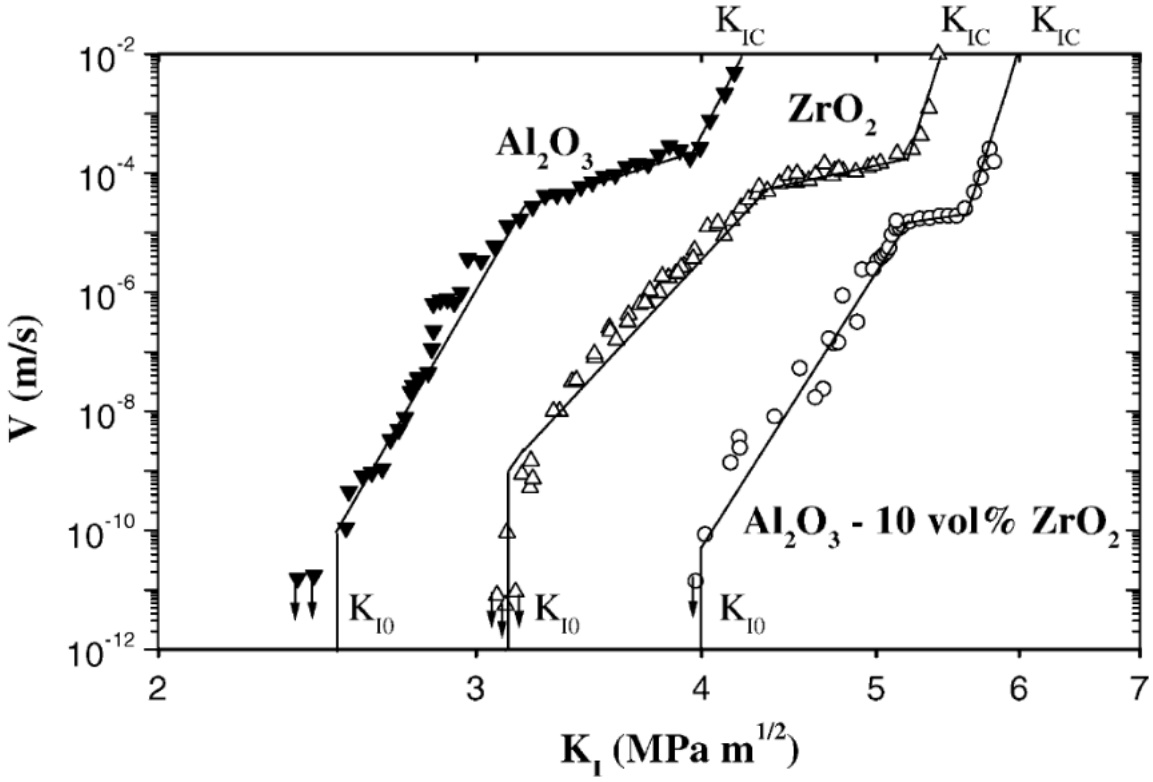


Figure 5-2-5. Relationship between crack propagation velocity (V) and stress intensity factor (KI) for Al_2O_3 (▼), ZrO_2 (△), and ZTA (10 vol% ZrO_2) (○) nanocomposites. Arrows indicate that no crack propagation was observed during testing. (Original figure from the paper)

The experimental results indicate that the Al_2O_3–10 vol% ZrO_2 nanocomposite ceramic exhibits the best crack propagation resistance among the three bioceramics. Its K_{IC} value reaches$(5.9 \pm 0.2)MPa \cdot m^{1/2}$, which is higher than that of single-phase alumina and slightly higher than that of single-phase zirconia. More importantly, the K_{I0} of this ZTA composite ceramic reached $(4.0 \pm 0.2)MPa \cdot m^{(1/2)}$, whereas that of single-phase alumina and zirconia was only $(2.5 \pm 0.2)and(3.1 \pm 0.2)MPa \cdot m^{(1/2)}$, respectively. This indicates that ZTA not only exhibits higher resistance to rapid fracture but also demonstrates greater resistance to slow crack propagation. This performance enhancement primarily stems from the synergistic interaction between the alumina matrix and the phase-transforming tetragonal ZrO_2 particles. During crack propagation, the tetragonal ZrO_2 phase undergoes a stress-induced transformation from the tetragonal to the monoclinic phase, accompanied by local volume expansion and the generation of compressive stress near the crack tip, thereby hindering further crack propagation. Therefore, compared to single-phase alumina, ZTA is more

suitable for load-bearing components in artificial joints that require long-term fracture reliability.

Therefore, flexural strength, Weibull statistics, fracture toughness, and FEA contact stress collectively validate the performance requirements proposed in Task 1, which stem from the design of the retained t-ZrO_2 phase stability in Task 3.

5.3 Standardised Product Testing

The ceramic components in artificial hip joints do not function as simple bar-shaped specimens, but rather work in conjunction with the femoral head or liner. Therefore, this section simulates the actual device environment and focuses on evaluating whether the finished femoral head can continue to function safely under conditions of assembly, cyclic loading, hydrothermal aging, and contact stress at the tapered connection.

The geometric shape of the sample is shown in the figure below, and the specific test details are shown in the table below.



Figure 5-3-1. The geometric shape of the sample(36mm).

Table 5-3-1. Quality Assurance Testing Standards and Acceptance Requirements for Ceramic Femoral Heads of Production Grade.

Testing/Standards	Content	Requirements
ISO 7206-2[60]	Sphericity, dimensions, and surface roughness of the joint surfaces	Ceramic femoral head joint surface: Ra,max ≤0.02 μm, Rt,max ≤1.0 μm; sphericity deviation ≤10 μm; femoral head diameter tolerance: −0.2 mm to 0 mm
ISO 7206-10[61]	Static Burst Strength Test of the Femoral Head	The average burst strength shall be >46 kN, and the strength of any single specimen shall not be less than 20 kN.
ISO 13356[62]	The Effects of Simulated Body Fluids, Hydrothermal Environments, and Cyclic Loading on the Stability and Strength of the ZTA Phase	High-pressure steam aging at 132 °C and 2 bar may be performed, combined with a cyclic loading of 10 kN; after aging, the burst strength shall still meet the limits specified in ISO 7206-10/FDA.
ISO 14242-1[63]	Wear of Simulated Femoral Heads and Liner-Liner Assemblies	Using 5 × 10^6 cycles, 1 Hz, 0.2–2 kN load, and 2 mm microseparation; the ZTA/ZTA

	Under Gait Cycles, Lubrication, and Edge Loading	pair should exhibit low wear rates and limited stripe wear

ISO 7206-10 is the most directly relevant standard, which uses a 36 mm ZTA ceramic femoral head and a lined design, loaded at a rate of 0.5 kN/s until burst failure. This method allows for the simultaneous evaluation of the ceramic body strength, the machining quality of the inner conical bore, the ceramic–metal contact stress, and assembly geometric errors. Additionally, the study incorporated hydrothermal aging at 132 °C and 2 bar, as well as cyclic loading at 10 kN, to simulate the body fluid environment and long-term mechanical loads.

Realistic sliding wear was evaluated in a 12-channel hip joint simulator using microseparation testing. This model simulates edge-to-edge contact between the femoral head and the cup rim during gait using a 2 mm separation, with a load range of 0.2–2 kN, a frequency of 1 Hz, and a total of 5×10^6 cycles.

ISO 13356 pertains to the evaluation of hydrothermal aging in zirconia-containing bioceramics. Compared to single-phase Y-TZP, ZTA has a lower volume fraction of ZrO_2, and the ZrO_2 is dispersed and isolated within the Al_2O_3 matrix; consequently, the continuous t→m phase transition is suppressed. This is one of the key reasons why ZTA is used in long-term implantable ceramic components.

The primary failure modes observed in service include slow crack growth, low-temperature hydrothermal aging, and edge contact damage caused by micro-separation. These failure modes can be controlled by adjusting the design parameters in Tasks 2–4: Y_2O_3-doped t-ZrO_2 in Task 2; the sintering phase stability window in Task 3 to control the retained tetragonal fraction; while densification, atmosphere control, and surface finish quality in Task 4 determine the final defect population and surface crack risk.

Therefore, Section 5.3 extends the validation of material-level performance to finished femoral heads, demonstrating that this ZTA system meets the device application requirements specified in Task 1, while also indicating that typical failure risks can be mitigated through doping, phase stability, sintering, and defect control as described in Tasks 2–4.

Task 6	Conclusions and outlook (10 marks)	
Suggested length	Indicative length 1000-1500 words	**Xuanming Che 、 Haotai Xu 、 Xiangjun Zheng、Yongqi Xu**

This report establishes a comprehensive design framework for ZTA ceramic femoral heads and liners in total hip replacement, covering service requirements, intrinsic material structure, phase stability, manufacturing processes, and final product validation. Task 1 begins by examining the actual in-service environment of ceramic hip joint bearings, establishing that the material must simultaneously meet requirements for high hardness, high wear resistance, sufficient fracture toughness, low particulate release, and long-term in vivo stability. This section provides evaluation criteria for all subsequent material selection and process design.

Task 2 explains the fundamental reasons why ZTA can meet the requirements outlined in Task 1 from the perspectives of chemical bonding, crystal structure, and oxide stability. The high bond energies, low solubility, and stable oxide lattices of Al–O, Zr–O, and Y–O bonds underpin ZTA's high hardness, chemical stability, and biocompatibility; meanwhile, the stress-induced t→m phase transformation of the metastable tetragonal ZrO_2 phase provides a key mechanism for enhancing fracture toughness.

Task 3 further translates the microstructural toughening mechanisms identified in Task 2 into engineering principles for phase design. Phase diagram and phase stability analyses indicate that the key to ZTA's performance lies not in simply increasing the ZrO_2 content or raising the sintering temperature, but in retaining appropriately sized, uniformly distributed, and metastable t-ZrO_2 toughening phases within the Al_2O_3 matrix. This conclusion validates the effectiveness of the phase transformation toughening mechanism identified in Task 2, while also imposing constraints on the processing route for Task 4: the process must achieve high densification without causing abnormal ZrO_2 grain growth, loss of phase composition control, or loss of metastability in the toughening phase.

Based on the performance targets and phase stability requirements derived from Tasks 1–3, Task 4 translates the material design into specific fabrication processes. This report selected high-purity α-Al_2O_3 and commercial 3Y-TZP powders, employing a process route involving wet ball milling, spray granulation, low-pressure precompaction, cold isostatic pressing, degreasing, two-step unpressurized sintering, and HIP post-treatment. This process sequence corresponds directly to the previous sections: wet ball milling and spray granulation serve the "uniform distribution of both phases" requirement in Task 3; CIP reduces the density gradient in the green body, addressing the requirements for reliability and low defect rates in Task 1; two-step sintering and HIP enhance density while suppressing abnormal grain growth, thereby maintaining the t-ZrO_2 toughening structure required for Tasks 2 and 3.

Task 5 constitutes the validation loop for the first four tasks. SEM/EDS is used to confirm whether Task 4 has truly achieved a uniform microstructure; XRD/Raman is used to verify the phase composition and t-ZrO_2 retention required by Task 3; density, hardness, flexural strength, fracture toughness, and Weibull statistics are used to evaluate the mechanical and reliability metrics defined in Task 1; FEA contact stress analysis, as well as finished-product-level tests such as ISO 7206-10 and ISO 14242, further link the laboratory material properties to the actual hip joint service environment.

Therefore, Task 5 is not merely a simple performance test, but rather a reverse validation to determine whether the requirements defined in Task 1, the mechanism assessment in Task 2, the phase design principles in Task 3, and the process route in Task 4 can collectively support the final device application.

Taken as a whole, Tasks 1–5 form a closed-loop process extending from application requirements to performance validation. Task 1 defines the requirements for strength, wear resistance, toughness, and long-term stability of ceramic hip bearings; Task 2 demonstrates that Al_2O_3 and ZrO_2 are complementary in terms of hardness, stability, and toughening mechanisms; Task 3 further indicates that this advantage depends on t-ZrO_2 of appropriate size and in a metastable state; Task 4 translates these phase structure requirements into powder mixing, forming, sintering, and HIP post-treatment; Task 5 verifies the validity of the design through microstructural analysis, phase composition, mechanical properties, and finished-product testing. Therefore, the performance of ZTA stems from the combined effects of material composition, phase structure, and processing control, rather than any single factor.

This report establishes a relatively comprehensive design rationale for ZTA ceramics used in hip joint femoral heads and liners, but some limitations remain. First, the material properties presented in the report are primarily based on literature data, redrawn figures, theoretical analysis, and FEA model inferences, and have not yet been verified using ZTA samples actually prepared by this group. Consequently, the conclusions regarding phase stability, t→m phase transformation toughening, and grain size control presented in Tasks 2 and 3 may still be influenced by powder purity, the dispersion state of ZrO_2, sintering temperature, and grain growth under actual processing conditions. Second, although the wet ball milling, CIP, two-step sintering, and HIP routes proposed in Task 4 can theoretically improve density and control microstructure, issues such as powder agglomeration, ball milling contamination, degreasing cracking, residual porosity, or polishing damage may still arise during actual preparation. These defects directly affect the strength, fracture toughness, and Weibull reliability results in Task 5; therefore, there is currently a lack of experimental evidence supporting the relationship between the current process route and the final performance.

Furthermore, the testing protocol and FEA contact stress analysis in Task 5 can only partially simulate the in-service environment of the hip joint. Actual implants will experience long-term cyclic loading, changes in lubrication conditions, micro-separation, assembly errors, and hydrothermal aging; these factors may simultaneously influence wear, phase transformation, and fatigue failure. The FEA modeling is an overly idealized two-dimensional axisymmetric model that assumes linearly elastic isotropic materials and ignores the effects of friction. The discussion in this report regarding long-term clinical failure data and hydrothermal aging literature remains insufficient. The feedback relationship between Task 4 and Task 5 could be further strengthened; for example, it should explain how the test results can be used to optimize sintering temperature, HIP parameters, and ZrO_2 content.

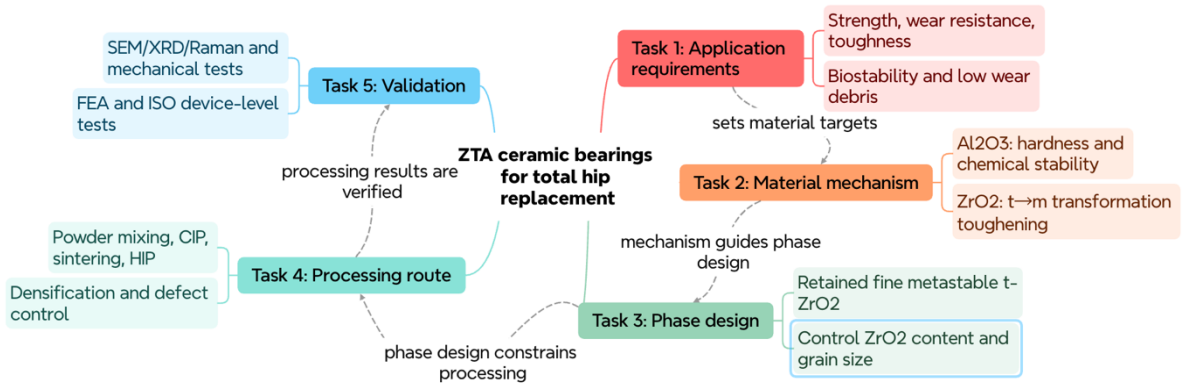


Figure 6-1-1. A summary diagram of the design logic for ZTA ceramic hip joint bearings, created using XMind.

Reference List

1. Piconi C, Maccauro G, Muratori F. Alumina and Zirconia Ceramics in Joint Replacements.

2. Hannouche D, Zaoui A, Zadegan F, Sedel L, Nizard R. Thirty years of experience with alumina-on-alumina bearings in total hip arthroplasty. International Orthopaedics (SICOT). 2011 Feb;35(2):207–13. doi:10.1007/s00264-010-1187-1

3. Orion Orthopaedic Surgery. Total Hip Replacement Surgery using Direct Anterior Approach (DAA). In: Orion Orthopaedic Surgery [Internet]. 2021. Available from: https://www.orionortho.sg/total-hip-replacement-using-direct-anterior-approach-daa

4. Boutin P. Arthroplastie totale de la hanche par prothèse en alumine frittée. Étude expérimentale et premières applications cliniques. Revue de Chirurgie Orthopédique et Traumatologique. 2014 Feb;100(1):14–21. doi:10.1016/j.rcot.2013.12.006

5. McEntire BJ, Bal BS, Rahaman MN, Chevalier J, Pezzotti G. Ceramics and ceramic coatings in orthopaedics. Journal of the European Ceramic Society. 2015 Dec;35(16):4327–69. doi:10.1016/j.jeurceramsoc.2015.07.034

6. Implants for surgery — Wear of total hip-joint prostheses — Part 1: Loading and displacement parameters for wear-testing machines and corresponding environmental conditions for test. Geneva, Switzerland; 2014.

7. Mak MM, Jin ZM. Analysis of contact mechanics in ceramic-on-ceramic hip joint replacements. Proc Inst Mech Eng H. 2002 Apr 1;216(4):231–6. doi:10.1243/09544110260138718

8. Bergmann G, Deuretzbacher G, Heller M, Graichen F, Rohlmann A, Strauss J, et al. Hip contact forces and gait patterns from routine activities. Journal of Biomechanics. 2001 Jul;34(7):859–71. doi:10.1016/S0021-9290(01)00040-9

9. ISO. Implants for surgery — Ceramic materials — Part 2: Composite materials based on a high-purity alumina matrix with zirconia reinforcement. ISO 6474-2:2019. 2019.

10. Bartolomé JF, Bruno G, DeAza AH. Neutron diffraction residual stress analysis of zirconia toughened alumina (ZTA) composites. Journal of the European Ceramic Society. 2008 Jan;28(9):1809–14. doi:10.1016/j.jeurceramsoc.2007.12.037

11. Sommer F, Landfried R, Kern F, Gadow R. Mechanical properties of zirconia toughened alumina with 10–24 vol.% 1Y-TZP reinforcement. Journal of the European Ceramic Society. 2012 Dec;32(16):4177–84. doi:10.1016/j.jeurceramsoc.2012.06.019

12. Achakri, Hassan; Ben-Shlomo, Yoav; Biant, Leela; et al. The National Joint Registry 22nd Annual Report 2025 [Internet]. London: National Joint Registry; 2024 Dec. Available from: https://www.ncbi.nlm.nih.gov/books/NBK618752/

13. Ashby MF, Gibson LJ, Wegst U, Olive R. The mechanical properties of natural materials. I. Material property charts. Proceedings of the Royal Society of London Series A: Mathematical and Physical Sciences. 1995 Jul 8;450(1938):123–40. doi:10.1098/rspa.1995.0075

14. Lewis G. Properties of acrylic bone cement: State of the art review. J Biomed Mater Res. 1997;38(2):155–82. doi:10.1002/(SICI)1097-4636(199722)38:2<155::AID-JBM10>3.0.CO;2-C

15. Pruitt LA. Deformation, yielding, fracture and fatigue behavior of conventional and highly cross-linked ultra high molecular weight polyethylene. Biomaterials. 2005 Mar;26(8):905–15. doi:10.1016/j.biomaterials.2004.03.022

16. Nalla RK, Stölken JS, Kinney JH, Ritchie RO. Fracture in human cortical bone: local fracture criteria and toughening mechanisms. Journal of Biomechanics. 2005 Jul;38(7):1517–25. doi:10.1016/j.jbiomech.2004.07.010

17. Kurtz SM, Pruitt L, Jewett CW, Paul Crawford R, Crane DJ, Edidin AA. The yielding, plastic flow, and fracture behavior of ultra-high molecular weight polyethylene used in total joint replacements. Biomaterials. 1998 Nov;19(21):1989–2003. doi:10.1016/S0142-9612(98)00112-4

18. Morgan EF, Yetkinler DN, Constantz BR, Dauskardt RH. Mechanical properties of carbonated apatite bone mineral substitute: strength, fracture and fatigue behaviour. Journal of Materials Science: Materials in Medicine. 1997 Sep;8(9):559–70. doi:10.1023/A:1018550831834

19. Hench LL. Bioceramics: From Concept to Clinic. Journal of the American Ceramic Society. 1991 Jul;74(7):1487–510. doi:10.1111/j.1151-2916.1991.tb07132.x

20. Martin RB, Burr DB, Sharkey NA. Skeletal Tissue Mechanics [Internet]. New York, NY: Springer New York; 1998 [cited 2026 May 28]. Available from: http://link.springer.com/10.1007/978-1-4757-2968-9 doi:10.1007/978-1-4757-2968-9

21. None Available. Materials Data on Al2O3 by Materials Project [Internet]. LBNL Materials Project; Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA (United States); 2020 [cited 2026 May 28]. Available from: https://www.osti.gov/servlets/purl/1187823/ doi:10.17188/1187823

22. Fathalian M, Postek E, Tahani M, Sadowski T. A Comprehensive Study of Al2O3 Mechanical Behavior Using Density Functional Theory and Molecular Dynamics. Molecules. 2024 Mar 5;29(5):1165. doi:10.3390/molecules29051165

23. None Available. Materials Data on ZrO2 by Materials Project [Internet]. LBNL Materials Project; Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA (United States); 2020 [cited 2026 May 28]. Available from: https://www.osti.gov/servlets/purl/1200956/ doi:10.17188/1200956

24. Fabris S. A stabilization mechanism of zirconia based on oxygen vacancies only. Acta Materialia. 2002 Dec 3;50(20):5171–8. doi:10.1016/S1359-6454(02)00385-3

25. Stapper G, Bernasconi M, Nicoloso N, Parrinello M. ***Ab initio*** study of structural and electronic properties of yttria-stabilized cubic zirconia. Phys Rev B. 1999 Jan 1;59(2):797–810. doi:10.1103/PhysRevB.59.797

27. Lange FF. Transformation toughening.

28. Subhoni M, Kholmurodov K, Doroshkevich A, Asgerov E, Yamamoto T, Lyubchyk A, et al. Density functional theory calculations of the water interactions with ZrO_2

nanoparticles Y_2 O_3 doped. J Phys: Conf Ser. 2018 Mar;994:012013. doi:10.1088/1742-6596/994/1/012013

29. None Available. Materials Data on ZrO2 by Materials Project [Internet]. LBNL Materials Project; Lawrence Berkeley National Laboratory (LBNL), Berkeley, CA (United States); 2020 [cited 2026 May 28]. Available from: https://www.osti.gov/servlets/purl/1202679/ doi:10.17188/1202679

30. Morse PM. Diatomic Molecules According to the Wave Mechanics. II. Vibrational Levels. Phys Rev. 1929 Jul 1;34(1):57–64. doi:10.1103/PhysRev.34.57

31. Luo YR. Comprehensive Handbook of Chemical Bond Energies [Internet]. 1st edn. CRC Press; 2007 [cited 2026 May 28]. Available from: https://www.taylorfrancis.com/books/9781420007282 doi:10.1201/9781420007282

32. Ackermann RJ, Rauh EG, Thorn RJ. Thermodynamic Properties of Gaseous Yttrium Monoxide. Correlation of Bonding in Group III Transition-Metal Monoxides. The Journal of Chemical Physics. 1964 Feb 1;40(3):883–9. doi:10.1063/1.1725221

33. Girifalco LA, Weizer VG. Application of the Morse Potential Function to Cubic Metals. Phys Rev. 1959 May 1;114(3):687–90. doi:10.1103/PhysRev.114.687

34. Chevalier Je, Grandjean S, Kuntz M, Pezzotti G. On the kinetics and impact of tetragonal to monoclinic transformation in an alumina/zirconia composite for arthroplasty applications.

35. Zhu Yong, Li Juan. Preparation and Forming Processes of Medical-Grade Yttrium Oxide-Stabilized Zirconia Ceramics. Chinese Ceramics. 2020;56(5):16–21. doi:10.16521/j.cnki.issn.1001-9642.2020.05.003

36. Lakiza SN, Lopato LM, Shevchenko AV. METHODS OF INVESTIGATION OF PROPERTIES OF POWDER MATERIALS.

37. Harjo S, Motohashi Y, Kojima N. Phase Transformation and Domain Switching Behaviors during Superplastic Deformation in 3Y-TZP. J Ceram Soc Japan. 2003;111(1297):680–7. doi:10.2109/jcersj.111.680

38. Lakiza SN, Lopato LM, Nazarenko LV, Zaitseva ZA. The liquidus surface in the Al2O3-ZrO2-Y2O3 phase diagram. Powder Metall Met Ceram. 1995;33(11–12):595–8. doi:10.1007/BF00559667

39. Wang Daning, Guo Yongquan, Liang Kaiming, Tao Kun. Study of the Crystal Structure of Zirconia Using the Rietveld Method. Science in China (Series A). 1998;(9):823–9.

40. Rittidech A, Suekwamsue N. Influence of yttria additives on structural, microstructural and mechanical properties of alumina–zirconia composites prepared by two-stage sintering. Ceramics International. 2015 Jul 1;The 9th Asian Meeting on Electroceramics (AMEC-9)41:S123–6. doi:10.1016/j.ceramint.2015.03.118

41. Wang L. Design and optimization of coating structure for the thermal barrier coatings fabricated by atmospheric plasma spraying via finite element method.

42. Park NK, Choi HY, Kim DH, Lee TJ, Kang M, Lee WG, et al. Purification of Al(OH)3 synthesized by Bayer process for preparation of high purity alumina as sapphire raw

material. Journal of Crystal Growth. 2013 Jun;373:88–91. doi:10.1016/j.jcrysgro.2012.12.004

43. Zhao Y, Deng J, Li W, Liu J, Yuan W. Grain Growth Behavior of Alumina in Zirconia-Toughened Alumina (ZTA) Ceramics During Pressureless Sintering. Crystals. 2025 Jan 17;15(1):89. doi:10.3390/cryst15010089

44. Lange FF. Transformation toughening: Part 4 Fabrication, fracture toughness and strength of Al2O3-ZrO2 composites. J Mater Sci. 1982 Jan;17(1):247–54. doi:10.1007/BF00809060

45. Yin S, Sun C, Li X, Zhang Y, Fang X, Ma H, et al. Phase composition, microstructure, and mechanical properties of ZTA ceramics prepared by gelcasting using multi-gradation spherical powders. Journal of Alloys and Compounds. 2025 Feb 25;1017:178984. doi:10.1016/j.jallcom.2025.178984

46. DING Kaidong. Recent Advances in Zirconia-Toughened Alumina Ceramics. Materials Engineering. 2025;53(4):75. doi:10.11868/j.issn.1001-4381.2024.000620

47. Zhang B, Lin X, Yan X, Luo J, Li Z, Cheng B, et al. Microwave-sintered graphene/ZTA biocomposites for artificial joints: Synergistic mechanical reinforcement, tribological enhancement, and biofunctional improvement. Journal of Materials Research and Technology. 2026 May 1;42:4872–88. doi:10.1016/j.jmrt.2026.04.129

48. Sequeira S, Fernandes MH, Neves N, Almeida MM. Development and characterization of zirconia–alumina composites for orthopedic implants. Ceramics International. 2017 Jan;43(1):693–703. doi:10.1016/j.ceramint.2016.09.216

49. Lewis JA. Colloidal Processing of Ceramics. Journal of the American Ceramic Society. 2000 Oct;83(10):2341–59. doi:10.1111/j.1151-2916.2000.tb01560.x

50. De Aza AH, Chevalier J, Fantozzi G, Schehl M, Torrecillas R. Crack growth resistance of alumina, zirconia and zirconia toughened alumina ceramics for joint prostheses. Biomaterials. 2002 Feb;23(3):937–45. doi:10.1016/S0142-9612(01)00206-X

51. Li J, Cai Q, Luo G, Zhong X, Shen Q, Tu R, et al. Zirconia Toughened Alumina Ceramics via Forming Intragranular Structure. Materials. 2024 Mar 12;17(6):1309. doi:10.3390/ma17061309

52. Chen Y, Fan B, Shao G, Zhang R. Preparation of large size ZTA ceramics with eccentric circle shape by microwave sintering. J Adv Ceram. 2016 Dec;5(4):291–7. doi:10.1007/s40145-016-0202-4

53. Standard Reference Material 660c: Line Position and Line Shape Standard for Powder Diffraction (Lanthanum Hexaboride Powder). Gaithersburg, MD: National Institute of Standards and Technology; 2015 Oct. Report SRM 660c.

54. Standard Guide for Quantitative Analysis by Energy-Dispersive Spectroscopy. West Conshohocken, PA; 2019.

55. Microbeam analysis — Scanning electron microscopy — Guidelines for calibrating image magnification. Geneva, Switzerland; 2016.

56. Tang D, Lim HB, Lee KJ, Lee CH, Cho WS. Evaluation of mechanical reliability of zirconia-toughened alumina composites for dental implants. Ceramics International. 2012 Apr;38(3):2429–36. doi:10.1016/j.ceramint.2011.11.009

57. Danzer R, Supancic P, Pascual J, Lube T. Fracture statistics of ceramics – Weibull statistics and deviations from Weibull statistics. Engineering Fracture Mechanics. 2007 Dec;74(18):2919–32. doi:10.1016/j.engfracmech.2006.05.028

58. Tateiwa T, Marin E, Rondinella A, Ciniglio M, Zhu W, Affatato S, et al. Burst Strength of BIOLOX®delta Femoral Heads and Its Dependence on Low-Temperature Environmental Degradation. Materials. 2020 Jan 12;13(2):350. doi:10.3390/ma13020350

59. Horgan CO, Knowles JK. Recent Developments Concerning Saint-Venant's Principle. In: Advances in Applied Mechanics [Internet]. Elsevier; 1983 [cited 2026 May 28]. p. 179–269. Available from: https://linkinghub.elsevier.com/retrieve/pii/S0065215608702448 doi:10.1016/S0065-2156(08)70244-8

60. Implants for surgery — Partial and total hip joint prostheses — Part 2: Articulating surfaces made of metallic, ceramic and plastics materials. Geneva, Switzerland; 2011.

61. Implants for surgery — Partial and total hip-joint prostheses — Part 10: Determination of resistance to static load of modular femoral heads. Geneva, Switzerland; 2018.

62. Implants for surgery — Ceramic materials based on yttria-stabilized tetragonal zirconia (Y-TZP). 2015.

63. Implants for surgery — Wear of total hip-joint prostheses — Part 1: Loading and displacement parameters for wear-testing machines and corresponding environmental conditions for test. 2014.

Authors contribution statement (to be included in this report)

Xuanming Che served as the group leader and was mainly responsible for overall work planning, task allocation, organising group meetings, and coordinating progress. During the preparation of the report, he clarified the main storyline of the report and ensured that Tasks 1–6 formed a coherent argument from application requirements and material selection to crystal structure, phase stability, processing route, characterisation, and performance validation. Xuanming Che also integrated the sections submitted by group members and revised the overall structure, language, figure and table numbering, reference format, and final layout of the report.

Haotai Xu, Xiangjun Zheng, and Yongqi Xu were responsible for completing their assigned sections, including literature review, information organisation, figure preparation, technical analysis, and writing. All members participated in group discussions at different stages and revised their sections based on feedback.

Haotai Xu and Xiangjun Zheng also contributed to organising part of the report logic, helping improve the connections between sections and making the argument across different tasks more coherent. Haotai Xu and Xuanming Che jointly discussed the microstructure of ZTA ceramics and determined an appropriate grain-size range for this report based on literature and material performance requirements. This provided a basis for later discussions of transformation toughening, sintering control, and performance validation.

In addition, Xuanming Che helped Yongqi Xu write and revise his assigned text, including improving paragraph logic, technical expression, and language consistency, and helping integrate the relevant content into the overall storyline of the report. Yongqi Xu confirmed the technical content of his section and participated in the final revision.

The final report was reviewed by all group members. All authors confirm that they are familiar with the overall content of the report and take responsibility for their own contributions and the final submitted version.

Declaration on the use of AI (to be included in this report)

During the preparation of this report, the authors used OpenAI ChatGPT/Codex as an AI-assisted tool. The use of AI was mainly limited to language improvement, text organisation, information organisation, and technical assistance, with the purpose of improving the readability, logical coherence, and formatting consistency of the report. Specifically, the AI tool was used to assist with Chinese-English expression, paragraph polishing, sentence structure improvement, organisation of logical connections between sections, and checking figure titles, table descriptions, and reference formatting.

In addition, the AI tool was used to assist with writing and modifying some Python/JavaScript scripts. These scripts were mainly used for data processing and visualisation in this report, including extracting data points from published literature or public datasets, redrawing NJR data figures, organising material property data for the Ashby chart, assisting quantitative analysis of particle size and phase distribution in SEM images, assisting XRD pattern digitisation and phase-composition trend analysis, conducting Weibull reliability-related calculations, and generating or adjusting some figures and intermediate data files. These tasks were all based on published literature, course materials, original images, tables, or data files compiled by the authors.

The AI tool was also used to assist with reference and source management, such as checking DOIs, organising the reference list, helping generate reference download records, and identifying data sources that required further verification. For all figures, tables, data, and conclusions used in this report, the authors manually reviewed and revised them against the original literature, course requirements, source data, calculation outputs, and image results.

It should be noted that the AI tool was not used to fabricate experimental data, invent references, create false results, replace the authors' understanding of the material system, or substitute the authors' final academic judgement. The content of this report concerning ZTA ceramic material selection, crystal structure, phase stability, processing route, SEM/XRD characterisation, mechanical properties, Weibull reliability analysis, FEA contact-stress interpretation, and final conclusions was completed by the authors based on literature review, data organisation, and technical discussion. AI outputs were used only as auxiliary drafts, language-improvement support, code assistance, and information-organisation tools.

After using the AI tool, the authors checked, revised, deleted, and integrated the relevant content as necessary. The authors take full responsibility for the final content of the report, data interpretation, figure and table presentation, reference accuracy, and academic integrity.